

YangPeng33235422

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R Markdown

Question 1

```
# Setup
#
set.seed(9999)
library(ggplot2)
library(tidyr)
library(knitr)
library(kableExtra)
library(gt)

VC <- read.csv("/Users/allison.py/Desktop/Y4/s1/FIT3152/FIT3152_Assignment1/FIT3152A1Data_YangPeng.csv")

theme_pub <- theme_minimal(base_size = 12, base_family = "serif") +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5,
                               margin = margin(b = 10)),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "#333333"),
    axis.line = element_line(color = "#333333", linewidth = 0.4),
    axis.ticks = element_line(color = "#333333", linewidth = 0.3),
    panel.grid.major = element_line(color = "grey88", linewidth = 0.3),
    panel.grid.minor = element_blank(),
    legend.position = "bottom",
    legend.title = element_text(size = 10),
    legend.text = element_text(size = 9),
    legend.key.size = unit(0.8, "lines"),
    plot.margin = margin(12, 15, 8, 10),
    strip.text = element_text(size = 11, face = "bold")
  )

# Q1 - Data

# Basic structure
str(VC)
```

```
## 'data.frame':   100000 obs. of  36 variables:
## $ Wave          : int  7 3 6 1 6 6 7 7 3 6 ...
## $ Country       : chr  "CZE" "BRA" "ZAF" "MEX" ...
```

```
## $ Year      : int 2022 1997 2013 1981 2012 2012 2017 2023 1996 2013 ...
## $ ILFam     : int 1 1 1 -4 1 1 1 1 2 1 ...
## $ ILPolitics : int 2 1 1 -4 3 3 3 2 3 2 ...
## $ ICQIndependence : int 0 0 0 1 1 1 1 1 1 1 ...
## $ ICQHardWork : int 0 1 1 1 1 1 0 1 0 0 ...
## $ ICQResonsibility: int 0 1 1 0 1 1 1 0 0 1 ...
## $ ICQDetermination: int 0 0 0 0 1 0 0 1 0 1 ...
## $ ICQFaith    : int 0 1 1 0 1 0 0 1 0 0 ...
## $ ICQUnselfishness: int 0 0 0 0 1 0 1 0 0 0 ...
## $ ACTReligion : int 0 2 0 0 1 0 2 1 1 1 ...
## $ ACTSport    : int 0 2 0 -4 0 1 0 0 1 1 ...
## $ ACTArtsEd   : int 0 0 2 0 0 0 0 1 0 0 ...
## $ ACTUnions   : int 0 1 1 0 0 0 0 1 1 0 ...
## $ Happy       : int 2 1 2 1 1 2 3 2 2 1 ...
## $ Health      : int 3 2 4 3 1 3 3 3 1 1 ...
## $ Trusted     : int 2 2 2 2 2 1 2 2 1 2 ...
## $ LifeSatis   : int 7 10 8 10 8 2 5 2 7 8 ...
## $ FutureRespect : int 2 1 -1 1 3 1 1 2 3 3 ...
## $ PolScale    : int 6 8 5 10 5 5 3 3 7 3 ...
## $ PrivateState : int 5 4 5 -4 7 5 7 6 5 4 ...
## $ Competition : int 4 4 7 -4 5 7 5 6 2 4 ...
## $ HardWork    : int 6 4 6 -4 1 8 4 4 3 5 ...
## $ PolLeader   : int 3 2 -1 -4 4 3 1 2 4 4 ...
## $ PolExperts  : int 2 2 3 -4 4 3 2 2 3 2 ...
## $ PolArmy     : int 3 4 2 -4 4 3 3 3 4 4 ...
## $ PolDemoc    : int 3 2 3 -4 1 2 1 3 1 2 ...
## $ CArmedForces : int 3 2 2 3 1 2 3 1 2 3 ...
## $ CPress      : int 3 3 2 4 1 4 4 2 3 3 ...
## $ CUnions     : int 3 3 3 4 3 3 2 2 2 3 ...
## $ CTelevision : int 2 2 1 -4 1 4 4 2 3 3 ...
## $ CGovernment : int 2 2 2 -4 1 3 3 4 2 3 ...
## $ CWomensMvt  : int 3 2 4 -4 3 3 2 2 2 2 ...
## $ CCourts     : int 2 3 1 3 1 2 2 3 2 2 ...
## $ CEU         : int 2 -4 -4 -4 -4 -4 3 -4 2 3 ...
```

```
colnames(VC)
```

```
## [1] "Wave"           "Country"        "Year"           "ILFam"
## [5] "ILPolitics"     "ICQIndependence" "ICQHardWork"    "ICQResonsibility"
## [9] "ICQDetermination" "ICQFaith"       "ICQUnselfishness" "ACTReligion"
## [13] "ACTSport"       "ACTArtsEd"      "ACTUnions"      "Happy"
## [17] "Health"         "Trusted"        "LifeSatis"      "FutureRespect"
## [21] "PolScale"       "PrivateState"   "Competition"    "HardWork"
## [25] "PolLeader"      "PolExperts"     "PolArmy"        "PolDemoc"
## [29] "CArmedForces"   "CPress"         "CUnions"        "CTelevision"
## [33] "CGovernment"    "CWomensMvt"     "CCourts"        "CEU"
```

```
sapply(VC, class)
```

```
##           Wave           Country           Year           ILFam
##      "integer"      "character"      "integer"      "integer"
##      ILPolitics ICQIndependence      ICQHardWork ICQResonsibility
##      "integer"           "integer"           "integer"           "integer"
```

```
## ICQDetermination      ICQFaith ICQUnselfishness      ACTReligion
##      "integer"         "integer"      "integer"         "integer"
##      ACTSport          ACTArtsEd      ACTUnions           Happy
##      "integer"         "integer"      "integer"         "integer"
##      Health            Trusted        LifeSatis         FutureRespect
##      "integer"         "integer"      "integer"         "integer"
##      PolScale          PrivateState    Competition        HardWork
##      "integer"         "integer"      "integer"         "integer"
##      PolLeader          PolExperts      PolArmy            PolDemoc
##      "integer"         "integer"      "integer"         "integer"
##      CArmedForces        CPress        CUnions            CTelevision
##      "integer"         "integer"      "integer"         "integer"
##      CGovernment        CWomensMvt      CCourts            CEU
##      "integer"         "integer"      "integer"         "integer"
```

```
dim(VC)
```

```
## [1] 100000      36
```

```
summary(VC)
```

```
##      Wave      Country      Year      ILFam
## Min.   :1.000   Length:100000   Min.   :1981   Min.   : -5.000
## 1st Qu.:3.000   Class :character   1st Qu.:1998   1st Qu.: 1.000
## Median :5.000   Mode  :character   Median :2006   Median : 1.000
## Mean   :4.782                      Mean   :2006   Mean   : 0.897
## 3rd Qu.:6.000                      3rd Qu.:2013   3rd Qu.: 1.000
## Max.   :7.000                      Max.   :2023   Max.   : 4.000
## ILPolitics      ICQIndependence      ICQHardWork      ICQResonsibility
## Min.   : -5.000   Min.   : -5.0000    Min.   : -5.000   Min.   : -5.0000
## 1st Qu.: 2.000   1st Qu.: 0.0000    1st Qu.: 0.000   1st Qu.: 0.0000
## Median : 3.000   Median : 0.0000    Median : 1.000   Median : 1.0000
## Mean   : 2.305   Mean   : 0.4368    Mean   : 0.479   Mean   : 0.6695
## 3rd Qu.: 3.000   3rd Qu.: 1.0000    3rd Qu.: 1.000   3rd Qu.: 1.0000
## Max.   : 4.000   Max.   : 1.0000    Max.   : 1.000   Max.   : 1.0000
## ICQDetermination      ICQFaith      ICQUnselfishness      ACTReligion
## Min.   : -5.0000    Min.   : -5.0000    Min.   : -5.0000    Min.   : -5.0000
## 1st Qu.: 0.0000    1st Qu.: 0.0000    1st Qu.: 0.0000    1st Qu.: 0.0000
## Median : 0.0000    Median : 0.0000    Median : 0.0000    Median : 0.0000
## Mean   : 0.2319    Mean   : 0.3219    Mean   : 0.2238    Mean   : -0.5423
## 3rd Qu.: 1.0000    3rd Qu.: 1.0000    3rd Qu.: 1.0000    3rd Qu.: 1.0000
## Max.   : 1.0000    Max.   : 1.0000    Max.   : 1.0000    Max.   : 2.0000
## ACTSport          ACTArtsEd      ACTUnions           Happy
## Min.   : -5.0000    Min.   : -5.000    Min.   : -5.0000    Min.   : -5.000
## 1st Qu.: -4.0000    1st Qu.: 0.000    1st Qu.: 0.0000    1st Qu.: 1.000
## Median : 0.0000    Median : 0.000    Median : 0.0000    Median : 2.000
## Mean   : -0.7475    Mean   : -0.729    Mean   : -0.7517    Mean   : 1.826
## 3rd Qu.: 0.0000    3rd Qu.: 0.000    3rd Qu.: 0.0000    3rd Qu.: 2.000
## Max.   : 2.0000    Max.   : 2.000    Max.   : 2.0000    Max.   : 4.000
## Health            Trusted        LifeSatis         FutureRespect
## Min.   : -5.00    Min.   : -4.000    Min.   : -5.000    Min.   : -4.000
## 1st Qu.: 2.00    1st Qu.: 1.000    1st Qu.: 5.000    1st Qu.: 1.000
## Median : 2.00    Median : 2.000    Median : 7.000    Median : 1.000
```

```
## Mean : 2.03 Mean : 1.582 Mean : 6.627 Mean : 1.316
## 3rd Qu.: 3.00 3rd Qu.: 2.000 3rd Qu.: 8.000 3rd Qu.: 2.000
## Max. : 5.00 Max. : 2.000 Max. :10.000 Max. : 3.000
## PolScale PrivateState Competition HardWork
## Min. :-5.000 Min. :-5.000 Min. :-5.000 Min. :-5.000
## 1st Qu.: -1.000 1st Qu.: 2.000 1st Qu.: 1.000 1st Qu.: 1.000
## Median : 5.000 Median : 5.000 Median : 3.000 Median : 3.000
## Mean : 3.305 Mean : 4.498 Mean : 2.725 Mean : 2.493
## 3rd Qu.: 6.000 3rd Qu.: 7.000 3rd Qu.: 5.000 3rd Qu.: 5.000
## Max. :10.000 Max. :10.000 Max. :10.000 Max. :10.000
## PolLeader PolExperts PolArmy PolDemoc
## Min. :-5.000 Min. :-5.00 Min. :-5.000 Min. :-5.0000
## 1st Qu.: 1.000 1st Qu.: 1.00 1st Qu.: 1.000 1st Qu.: 1.0000
## Median : 2.000 Median : 2.00 Median : 3.000 Median : 1.0000
## Mean : 1.671 Mean : 1.31 Mean : 1.851 Mean : 0.8509
## 3rd Qu.: 3.000 3rd Qu.: 3.00 3rd Qu.: 4.000 3rd Qu.: 2.0000
## Max. : 4.000 Max. : 4.00 Max. : 4.000 Max. : 4.0000
## CArmedForces CPress CUnions CTelevision
## Min. :-5.000 Min. :-5.000 Min. :-5.000 Min. :-5.000
## 1st Qu.: 1.000 1st Qu.: 2.000 1st Qu.: 2.000 1st Qu.: 2.000
## Median : 2.000 Median : 3.000 Median : 3.000 Median : 2.000
## Mean : 1.811 Mean : 2.279 Mean : 2.128 Mean : 1.873
## 3rd Qu.: 3.000 3rd Qu.: 3.000 3rd Qu.: 3.000 3rd Qu.: 3.000
## Max. : 4.000 Max. : 4.000 Max. : 4.000 Max. : 4.000
## CGovernment CWomensMvt CCourts CEU
## Min. :-5.000 Min. :-5.000 Min. :-5.000 Min. :-5.00
## 1st Qu.: 1.000 1st Qu.: 1.000 1st Qu.: 1.000 1st Qu.: -4.00
## Median : 2.000 Median : 2.000 Median : 2.000 Median : -4.00
## Mean : 1.779 Mean : 1.225 Mean : 1.217 Mean : -2.26
## 3rd Qu.: 3.000 3rd Qu.: 3.000 3rd Qu.: 3.000 3rd Qu.: -1.00
## Max. : 4.000 Max. : 4.000 Max. : 4.000 Max. : 4.00
```

SGP-specific check

```
spg_data <- unique(VC[VC$Country == "SGP", ])
summary(spg_data)
```

```
## Wave Country Year ILFam
## Min. :4.00 Length:1617 Min. :2002 Min. :-2.000
## 1st Qu.:4.00 Class :character 1st Qu.:2002 1st Qu.: 1.000
## Median :6.00 Mode :character Median :2012 Median : 1.000
## Mean :5.81 Mean :2012 Mean : 1.076
## 3rd Qu.:7.00 3rd Qu.:2020 3rd Qu.: 1.000
## Max. :7.00 Max. :2020 Max. : 4.000
## ILPolitics ICQIndependence ICQHardWork ICQResonsibility
## Min. :-2.000 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.: 2.000 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median : 3.000 Median :1.0000 Median :1.0000 Median :1.0000
## Mean : 2.511 Mean :0.6673 Mean :0.5813 Mean :0.7403
## 3rd Qu.: 3.000 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. : 4.000 Max. :1.0000 Max. :1.0000 Max. :1.0000
## ICQDetermination ICQFaith ICQUnselfishness ACTReligion
## Min. :0.0000 Min. :0.0000 Min. :0.0000 Min. :-4.0000
## 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.: -4.0000
## Median :0.0000 Median :0.0000 Median :0.0000 Median : 0.0000
```

```
## Mean :0.4249 Mean :0.2938 Mean :0.2777 Mean :-0.6555
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.: 1.0000
## Max. :1.0000 Max. :1.0000 Max. :1.0000 Max. : 2.0000
## ACTSport ACTArtsEd ACTUnions Happy
## Min. :-4.0000 Min. :-4.0000 Min. :-4.000 Min. :-2.000
## 1st Qu.: -4.0000 1st Qu.: -4.0000 1st Qu.: -4.000 1st Qu.: 1.000
## Median : 0.0000 Median : 0.0000 Median : 0.000 Median : 2.000
## Mean :-0.8596 Mean :-0.9153 Mean :-1.002 Mean : 1.754
## 3rd Qu.: 0.0000 3rd Qu.: 0.0000 3rd Qu.: 0.000 3rd Qu.: 2.000
## Max. : 2.0000 Max. : 2.0000 Max. : 2.000 Max. : 4.000
## Health Trusted LifeSatis FutureRespect
## Min. :-4.0000 Min. :-2.000 Min. :-2.00 Min. :-2.000
## 1st Qu.: -4.0000 1st Qu.: 1.000 1st Qu.: 6.00 1st Qu.: 1.000
## Median : 2.0000 Median : 2.000 Median : 7.00 Median : 2.000
## Mean : 0.3271 Mean : 1.669 Mean : 7.02 Mean : 1.626
## 3rd Qu.: 2.0000 3rd Qu.: 2.000 3rd Qu.: 8.00 3rd Qu.: 2.000
## Max. : 4.0000 Max. : 2.000 Max. :10.00 Max. : 3.000
## PolScale PrivateState Competition HardWork
## Min. :-4.0000 Min. :-5.000 Min. :-5.000 Min. :-5.000
## 1st Qu.: -4.0000 1st Qu.: 4.000 1st Qu.: 2.000 1st Qu.: -4.000
## Median : -4.0000 Median : 5.000 Median : 4.000 Median : 3.000
## Mean :-0.9858 Mean : 5.109 Mean : 3.869 Mean : 2.106
## 3rd Qu.: 3.0000 3rd Qu.: 6.000 3rd Qu.: 5.000 3rd Qu.: 5.000
## Max. :10.0000 Max. :10.000 Max. :10.000 Max. :10.000
## PolLeader PolExperts PolArmy PolDemoc
## Min. :-5.000 Min. :-5.000 Min. :-5.000 Min. :-5.000
## 1st Qu.: 2.000 1st Qu.: 2.000 1st Qu.: 3.000 1st Qu.: 1.000
## Median : 3.000 Median : 2.000 Median : 3.000 Median : 2.000
## Mean : 2.601 Mean : 2.344 Mean : 3.088 Mean : 1.636
## 3rd Qu.: 3.000 3rd Qu.: 3.000 3rd Qu.: 4.000 3rd Qu.: 2.000
## Max. : 4.000 Max. : 4.000 Max. : 4.000 Max. : 4.000
## CArmedForces CPress CUnions CTelevision
## Min. :-4.0000 Min. :-4.0000 Min. :-4.0000 Min. :-4.000
## 1st Qu.: -4.0000 1st Qu.: -4.0000 1st Qu.: -4.0000 1st Qu.: -4.000
## Median : 2.0000 Median : 2.0000 Median : 2.0000 Median : 2.000
## Mean : 0.2993 Mean : 0.5733 Mean : 0.5306 Mean : 0.572
## 3rd Qu.: 2.0000 3rd Qu.: 3.0000 3rd Qu.: 3.0000 3rd Qu.: 3.000
## Max. : 4.0000 Max. : 4.0000 Max. : 4.0000 Max. : 4.000
## CGovernment CWomensMvt CCourts CEU
## Min. :-4.0000 Min. :-5.0000 Min. :-4.0000 Min. :-4
## 1st Qu.: -4.0000 1st Qu.: -4.0000 1st Qu.: -4.0000 1st Qu.: -4
## Median : 2.0000 Median : 2.0000 Median : 2.0000 Median : -4
## Mean : 0.3092 Mean : 0.2721 Mean : 0.2851 Mean : -4
## 3rd Qu.: 2.0000 3rd Qu.: 2.0000 3rd Qu.: 2.0000 3rd Qu.: -4
## Max. : 4.0000 Max. : 4.0000 Max. : 4.0000 Max. : -4
```

```
sapply(spg_data[, sapply(spg_data, is.numeric)], function(x) sum(x < 0, na.rm = TRUE))
```

```
## Wave Year ILFam ILPolitics
## 0 0 1 3
## ICQIndependence ICQHardWork ICQResonsibility ICQDetermination
## 0 0 0 0
## ICQFaith ICQUnselfishness ACTReligion ACTSport
## 0 0 451 452
```

```
##      ACTArtsEd      ACTUnions      Happy      Health
##      451          452          1          451
##      Trusted      LifeSatis      FutureRespect      PolScale
##      10           3           15          1095
##      PrivateState      Competition      HardWork      PolLeader
##      8            6          453          74
##      PolExperts      PolArmy      PolDemoc      CArmedForces
##      86            45          52          459
##      CPress      CUnions      CTelevision      CGovernment
##      462          496          459          459
##      CWomensMvt      CCourts      CEU
##      566          467          1617
```

```
# Year and variable inspection
```

```
unique(VC[, "Year"])
```

```
## [1] 2022 1997 2013 1981 2012 2017 2023 1996 2001 2011 2010 1995 1991 2018 1982
## [16] 2007 2005 2009 1990 2006 2000 2021 2004 2002 2014 1998 2019 2020 1999 2008
## [31] 1984 2003 1989
```

```
unique(VC$Health)
```

```
## [1] 3 2 4 1 -4 -1 5 -2 -5
```

```
unique(VC$ICQIndependence)
```

```
## [1] 0 1 -4 -5 -2 -1
```

```
# Negative value counts per numeric column
```

```
num_cols <- names(VC)[sapply(VC, is.numeric)]
sapply(VC[num_cols], function(x) sum(x < 0, na.rm = TRUE))
```

```
##      Wave      Year      ILFam      ILPolitics
##      0          0      4316      6014
##      ICQIndependence      ICQHardWork      ICQResonsibility      ICQDetermination
##      788          1404          690          3040
##      ICQFaith      ICQUnselfishness      ACTReligion      ACTSport
##      1405          1826          24291          26012
##      ACTArtsEd      ACTUnions      Happy      Health
##      24368          23413          2138          2923
##      Trusted      LifeSatis      FutureRespect      PolScale
##      4109          1446          5873          29305
##      PrivateState      Competition      HardWork      PolLeader
##      12332          14653          22610          18514
##      PolExperts      PolArmy      PolDemoc      CArmedForces
##      20537          20573          17800          7375
##      CPress      CUnions      CTelevision      CGovernment
##      6195          12591          11240          13936
##      CWomensMvt      CCourts      CEU
##      22905          20054          75494
```

```
# Min / max / unique counts (excluding negatives)
```

```
sapply(VC[num_cols], function(x) {
  valid <- x[x >= 0]
  c(Min   = min(valid, na.rm = TRUE),
    Max   = max(valid, na.rm = TRUE),
    Unique = length(unique(valid)))
})
```

```
##      Wave Year ILFam ILPolitics ICQIndependence ICQHardWork ICQResonsibility
## Min      1 1981      1          1              0            0              0
## Max      7 2023      4          4              1            1              1
## Unique    7  33      4          4              2            2              2
##      ICQDetermination ICQFaith ICQUnselfishness ACTReligion ACTSport
## Min                0          0              0            0            0
## Max                1          1              1            2            2
## Unique             2          2              2            3            3
##      ACTArtsEd ACTUnions Happy Health Trusted LifeSatis FutureRespect
## Min              0          0          1          1          1          1          1
## Max              2          2          4          5          2          10         3
## Unique           3          3          4          5          2          10         3
##      PolScale PrivateState Competition HardWork PolLeader PolExperts PolArmy
## Min              1          1              1          1          1          1          1
## Max             10          10              10         10          4          4          4
## Unique           10          10              10         10          4          4          4
##      PolDemoc CArmedForces CPress CUnions CTelevision CGovernment CWomensMvt
## Min              1          1          1          1          1          1          1
## Max              4          4          4          4          4          4          4
## Unique           4          4          4          4          4          4          4
##      CCourts CEU
## Min              1          1
## Max              4          4
## Unique           4          4
```

```
# Unique value counts and values per column
```

```
sapply(VC, function(x) length(unique(x)))
```

```
##      Wave      Country      Year      ILFam
##      7          58          33          8
##      ILPolitics ICQIndependence ICQHardWork ICQResonsibility
##      8          6          6          6
##      ICQDetermination ICQFaith ICQUnselfishness ACTReligion
##      6          6          6          7
##      ACTSport      ACTArtsEd      ACTUnions      Happy
##      7          7          7          8
##      Health      Trusted      LifeSatis      FutureRespect
##      9          5          14          6
##      PolScale      PrivateState      Competition      HardWork
##      14          14          14          14
##      PolLeader      PolExperts      PolArmy      PolDemoc
##      8          8          8          8
##      CArmedForces      CPress      CUnions      CTelevision
##      8          8          8          8
##      CGovernment      CWomensMvt      CCourts      CEU
```

```
##           8           8           8           8
```

```
sapply(VC, function(x) sort(unique(x)))
```

```
## $Wave
## [1] 1 2 3 4 5 6 7
##
## $Country
## [1] "ARG" "ARM" "AUS" "BGD" "BLR" "BRA" "CAN" "CHE" "CHL" "CHN" "COL" "CYP"
## [13] "CZE" "DEU" "EGY" "ESP" "FIN" "GBR" "GEO" "HKG" "HUN" "IDN" "IND" "IRN"
## [25] "IRQ" "JOR" "JPN" "KGZ" "KOR" "MAR" "MDA" "MEX" "MYS" "NGA" "NLD" "NZL"
## [37] "PAK" "PER" "PHL" "POL" "PRI" "ROU" "RUS" "SGP" "SRB" "SVK" "SVN" "SWE"
## [49] "THA" "TUR" "TWN" "UKR" "URY" "USA" "VEN" "VNM" "ZAF" "ZWE"
##
## $Year
## [1] 1981 1982 1984 1989 1990 1991 1995 1996 1997 1998 1999 2000 2001 2002 2003
## [16] 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2017 2018 2019 2020
## [31] 2021 2022 2023
##
## $ILFam
## [1] -5 -4 -2 -1 1 2 3 4
##
## $ILPolitics
## [1] -5 -4 -2 -1 1 2 3 4
##
## $ICQIndependence
## [1] -5 -4 -2 -1 0 1
##
## $ICQHardWork
## [1] -5 -4 -2 -1 0 1
##
## $ICQResonsibility
## [1] -5 -4 -2 -1 0 1
##
## $ICQDetermination
## [1] -5 -4 -2 -1 0 1
##
## $ICQFaith
## [1] -5 -4 -2 -1 0 1
##
## $ICQUnselfishness
## [1] -5 -4 -2 -1 0 1
##
## $ACTReligion
## [1] -5 -4 -2 -1 0 1 2
##
## $ACTSport
## [1] -5 -4 -2 -1 0 1 2
##
## $ACTArtsEd
## [1] -5 -4 -2 -1 0 1 2
##
## $ACTUnions
## [1] -5 -4 -2 -1 0 1 2
```



```

##
## $Happy
## [1] -5 -4 -2 -1  1  2  3  4
##
## $Health
## [1] -5 -4 -2 -1  1  2  3  4  5
##
## $Trusted
## [1] -4 -2 -1  1  2
##
## $LifeSatis
## [1] -5 -4 -2 -1  1  2  3  4  5  6  7  8  9 10
##
## $FutureRespect
## [1] -4 -2 -1  1  2  3
##
## $PolScale
## [1] -5 -4 -2 -1  1  2  3  4  5  6  7  8  9 10
##
## $PrivateState
## [1] -5 -4 -2 -1  1  2  3  4  5  6  7  8  9 10
##
## $Competition
## [1] -5 -4 -2 -1  1  2  3  4  5  6  7  8  9 10
##
## $HardWork
## [1] -5 -4 -2 -1  1  2  3  4  5  6  7  8  9 10
##
## $PolLeader
## [1] -5 -4 -2 -1  1  2  3  4
##
## $PolExperts
## [1] -5 -4 -2 -1  1  2  3  4
##
## $PolArmy
## [1] -5 -4 -2 -1  1  2  3  4
##
## $PolDemoc
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CArmedForces
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CPress
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CUnions
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CTelevision
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CGovernment
## [1] -5 -4 -2 -1  1  2  3  4

```

```
##
## $CWomensMvt
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CCourts
## [1] -5 -4 -2 -1  1  2  3  4
##
## $CEU
## [1] -5 -4 -2 -1  1  2  3  4
```

```
# Country distribution
country_freq <- sort(table(VC$Country), decreasing = TRUE)
country_df <- data.frame(
  Country = names(country_freq),
  Count   = as.numeric(country_freq),
  Pct     = round(as.numeric(country_freq) / nrow(VC) * 100, 1)
)
head(country_df, 10) # Top 10
```

```
##      Country Count Pct
## 1      ZAF  4621 4.6
## 2      IND  4007 4.0
## 3      COL  3362 3.4
## 4      TUR  3290 3.3
## 5      MEX  3243 3.2
## 6      CHN  3025 3.0
## 7      RUS  2908 2.9
## 8      JPN  2587 2.6
## 9      USA  2522 2.5
## 10     EGY  2460 2.5
```

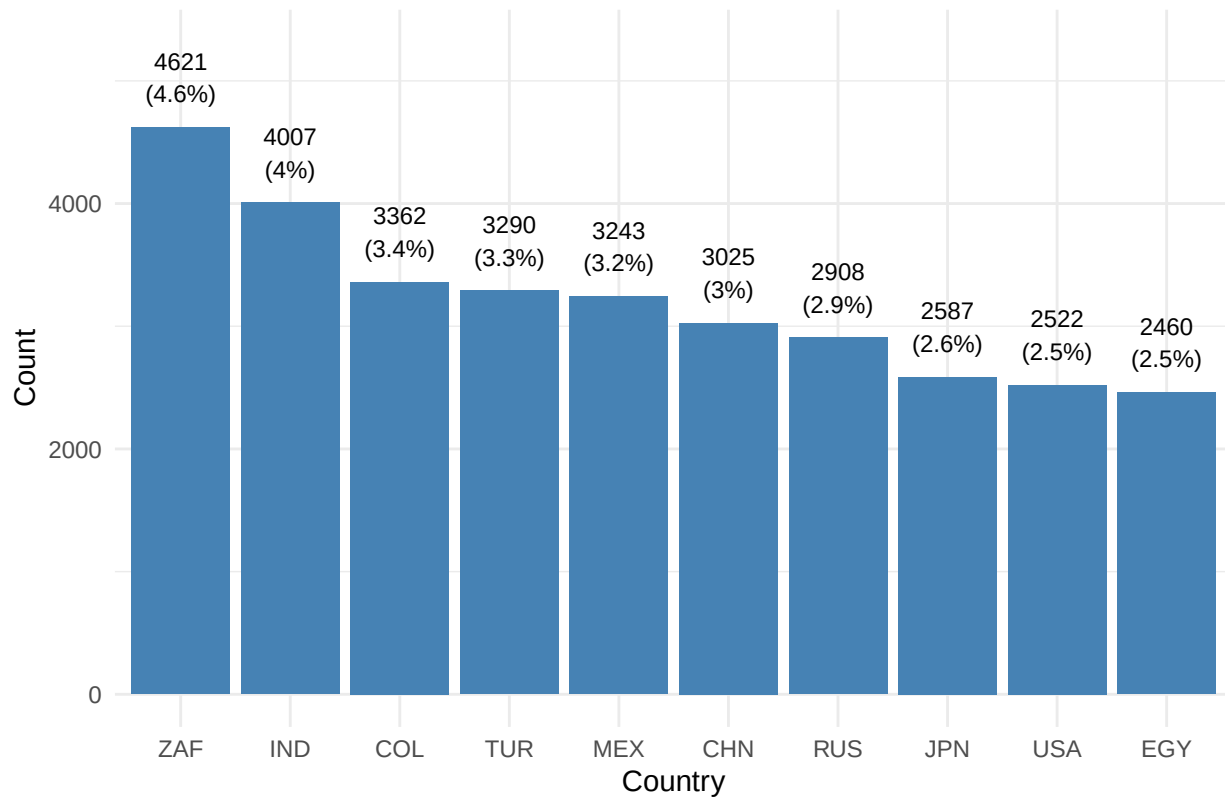
```
tail(country_df, 5) # Bottom 5
```

```
##      Country Count Pct
## 54     HUN   870 0.9
## 55     SVN   857 0.9
## 56     CYP   851 0.9
## 57     MDA   819 0.8
## 58     SVK   802 0.8
```

```
# Top-10 country bar chart
top10 <- head(country_df, 10)

ggplot(top10, aes(x = reorder(Country, -Count), y = Count)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = paste0(Count, "\n(", Pct, "%)")),
    vjust = -0.5, size = 3) +
  labs(title = "Top 10 Countries by Number of Respondents",
    x = "Country", y = "Count") +
  theme_minimal() +
  ylim(0, max(top10$Count) * 1.15)
```

Top 10 Countries by Number of Respondents



```
# =====
# Missing

# Identify
numeric_cols <- VC[, sapply(VC, is.numeric)]
negative_vals <- sort(unique(unlist(lapply(numeric_cols, function(col) col[col < 0]))))

# Count per missing code
neg1_counts <- sapply(VC, function(col) sum(col == -1, na.rm = TRUE))
neg2_counts <- sapply(VC, function(col) sum(col == -2, na.rm = TRUE))
neg4_counts <- sapply(VC, function(col) sum(col == -4, na.rm = TRUE))
neg5_counts <- sapply(VC, function(col) sum(col == -5, na.rm = TRUE))

sentinel_counts <- sapply(VC, function(col) sum(col %in% negative_vals))
pct_missing <- round(sentinel_counts / nrow(VC) * 100, 2)

# Missing / Sentinel Values

# Identify sentinel codes
numeric_cols <- VC[, sapply(VC, is.numeric)]
negative_vals <- sort(unique(unlist(lapply(numeric_cols, function(col) col[col < 0]))))

# Count per sentinel code
neg1_counts <- sapply(VC, function(col) sum(col == -1, na.rm = TRUE))
neg2_counts <- sapply(VC, function(col) sum(col == -2, na.rm = TRUE))
neg4_counts <- sapply(VC, function(col) sum(col == -4, na.rm = TRUE))
```

```

neg5_counts <- sapply(VC, function(col) sum(col == -5, na.rm = TRUE))

sentinel_counts <- sapply(VC, function(col) sum(col %in% negative_vals))
pct_missing <- round(sentinel_counts / nrow(VC) * 100, 2)

# Missing-value summary table
missing_summary <- data.frame(
  Variable = names(VC),
  Sentinel_neg4 = neg4_counts,
  Sentinel_neg1 = neg1_counts,
  Sentinel_neg2 = neg2_counts,
  Sentinel_neg5 = neg5_counts,
  Total_Missing = sentinel_counts,
  Pct_Missing = pct_missing,
  row.names = NULL
)
missing_summary <- missing_summary[order(-missing_summary$Pct_Missing), ]

missing_summary$Severity <- ifelse(
  missing_summary$Pct_Missing >= 20, "High (>=20%)",
  ifelse(missing_summary$Pct_Missing >= 10, "Moderate (10-20%)",
    ifelse(missing_summary$Pct_Missing >= 5, "Low (5-10%)",
      ifelse(missing_summary$Pct_Missing > 0, "Minimal (<5%)", "None"))))

missing_summary$Pct_NonResponse <- ifelse(
  missing_summary$Total_Missing > 0,
  round((missing_summary$Sentinel_neg1 + missing_summary$Sentinel_neg2) /
    missing_summary$Total_Missing * 100, 1),
  0)

# Replace sentinels with NA
VC_clean <- VC
for (v in names(VC_clean)) {
  if (is.numeric(VC_clean[[v]])) {
    VC_clean[[v]][VC_clean[[v]] %in% negative_vals] <- NA
  }
}

# Reverse confidence + wellbeing variables (make higher = more positive)
confidence_vars <- c("CArmedForces", "CPress", "CUnions", "CTelevision",
  "CGovernment", "CWomensMvt", "CCourts", "CEU")

VC_clean[, confidence_vars] <- lapply(VC_clean[, confidence_vars], function(x) 5 - x)
VC_clean$Happy <- 5 - VC_clean$Happy
VC_clean$Health <- 6 - VC_clean$Health
VC_clean$Trusted <- 3 - VC_clean$Trusted
VC_clean$FutureRespect <- 4 - VC_clean$FutureRespect

# Importance in life
il_vars <- c("ILFam", "ILPolitics")
VC_clean[il_vars] <- lapply(VC_clean[il_vars], function(x) 5 - x)

```

```

polsys_vars <- c("PolLeader", "PolExperts", "PolArmy", "PolDemoc")
VC_clean[polsys_vars] <- lapply(VC_clean[polsys_vars], function(x) 5 - x)

# Competition / HardWork

VC_clean$Competition <- 11 - VC_clean$Competition
VC_clean$HardWork <- 11 - VC_clean$HardWork

summary(VC_clean)

```

```

##      Wave      Country      Year      ILFam
##  Min.   :1.000   Length:100000   Min.   :1981   Min.   :1.000
##  1st Qu.:3.000   Class :character   1st Qu.:1998   1st Qu.:4.000
##  Median :5.000   Mode  :character   Median :2006   Median :4.000
##  Mean   :4.782                Mean   :2006   Mean   :3.889
##  3rd Qu.:6.000                3rd Qu.:2013   3rd Qu.:4.000
##  Max.   :7.000                Max.   :2023   Max.   :4.000
##                                     NA's   :4316
##      ILPolitics  ICQIndependence  ICQHardWork  ICQResponsibility
##  Min.   :1.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0
##  1st Qu.:2.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0
##  Median :2.000   Median :0.0000   Median :1.0000   Median :1.0
##  Mean   :2.348   Mean   :0.4674   Mean   :0.5405   Mean   :0.7
##  3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0
##  Max.   :4.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0
##  NA's   :6014   NA's   :788     NA's   :1404   NA's   :690
##      ICQDetermination  ICQFaith  ICQUnselfishness  ACTReligion
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000
##  Mean   :0.3601   Mean   :0.3784   Mean   :0.2976   Mean   :0.5499
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :2.0000
##  NA's   :3040   NA's   :1405   NA's   :1826   NA's   :24291
##      ACTSport  ACTArtsEd  ACTUnions  Happy
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :1.000
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:3.000
##  Median :0.0000   Median :0.0000   Median :0.0000   Median :3.000
##  Mean   :0.3765   Mean   :0.2752   Mean   :0.2182   Mean   :3.075
##  3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:4.000
##  Max.   :2.0000   Max.   :2.0000   Max.   :2.0000   Max.   :4.000
##  NA's   :26012   NA's   :24368   NA's   :23413   NA's   :2138
##      Health  Trusted  LifeSatis  FutureRespect
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:3.000   1st Qu.:1.000   1st Qu.:5.000   1st Qu.:2.000
##  Median :4.000   Median :1.000   Median :7.000   Median :3.000
##  Mean   :3.797   Mean   :1.272   Mean   :6.761   Mean   :2.463
##  3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:8.000   3rd Qu.:3.000
##  Max.   :5.000   Max.   :2.000   Max.   :10.000   Max.   :3.000
##  NA's   :2923   NA's   :4109   NA's   :1446   NA's   :5873
##      PolScale  PrivateState  Competition  HardWork
##  Min.   :1.000   Min.   :1.0   Min.   :1.000   Min.   :1.000

```

##	1st Qu.: 5.000	1st Qu.: 3.0	1st Qu.: 6.000	1st Qu.: 5.000
##	Median : 5.000	Median : 5.0	Median : 8.000	Median : 7.000
##	Mean : 5.694	Mean : 5.5	Mean : 7.233	Mean : 6.681
##	3rd Qu.: 7.000	3rd Qu.: 8.0	3rd Qu.:10.000	3rd Qu.: 9.000
##	Max. :10.000	Max. :10.0	Max. :10.000	Max. :10.000
##	NA's :29305	NA's :12332	NA's :14653	NA's :22610
##	PolLeader	PolExperts	PolArmy	PolDemoc
##	Min. :1.000	Min. :1.00	Min. :1.000	Min. :1.000
##	1st Qu.:1.000	1st Qu.:2.00	1st Qu.:1.000	1st Qu.:3.000
##	Median :2.000	Median :3.00	Median :2.000	Median :3.000
##	Mean :2.311	Mean :2.66	Mean :1.868	Mean :3.329
##	3rd Qu.:3.000	3rd Qu.:3.00	3rd Qu.:2.000	3rd Qu.:4.000
##	Max. :4.000	Max. :4.00	Max. :4.000	Max. :4.000
##	NA's :18514	NA's :20537	NA's :20573	NA's :17800
##	CArmedForces	CPress	CUnions	CTelevision
##	Min. :1.000	Min. :1.000	Min. :1.00	Min. :1.000
##	1st Qu.:2.000	1st Qu.:2.000	1st Qu.:2.00	1st Qu.:2.000
##	Median :3.000	Median :2.000	Median :2.00	Median :2.000
##	Mean :2.816	Mean :2.399	Mean :2.29	Mean :2.467
##	3rd Qu.:4.000	3rd Qu.:3.000	3rd Qu.:3.00	3rd Qu.:3.000
##	Max. :4.000	Max. :4.000	Max. :4.00	Max. :4.000
##	NA's :7375	NA's :6195	NA's :12591	NA's :11240
##	CGovernment	CWomensMvt	CCourts	CEU
##	Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
##	1st Qu.:2.000	1st Qu.:2.000	1st Qu.:2.000	1st Qu.:2.000
##	Median :2.000	Median :3.000	Median :3.000	Median :2.000
##	Mean :2.394	Mean :2.612	Mean :2.563	Mean :2.333
##	3rd Qu.:3.000	3rd Qu.:3.000	3rd Qu.:3.000	3rd Qu.:3.000
##	Max. :4.000	Max. :4.000	Max. :4.000	Max. :4.000
##	NA's :13936	NA's :22905	NA's :20054	NA's :75494

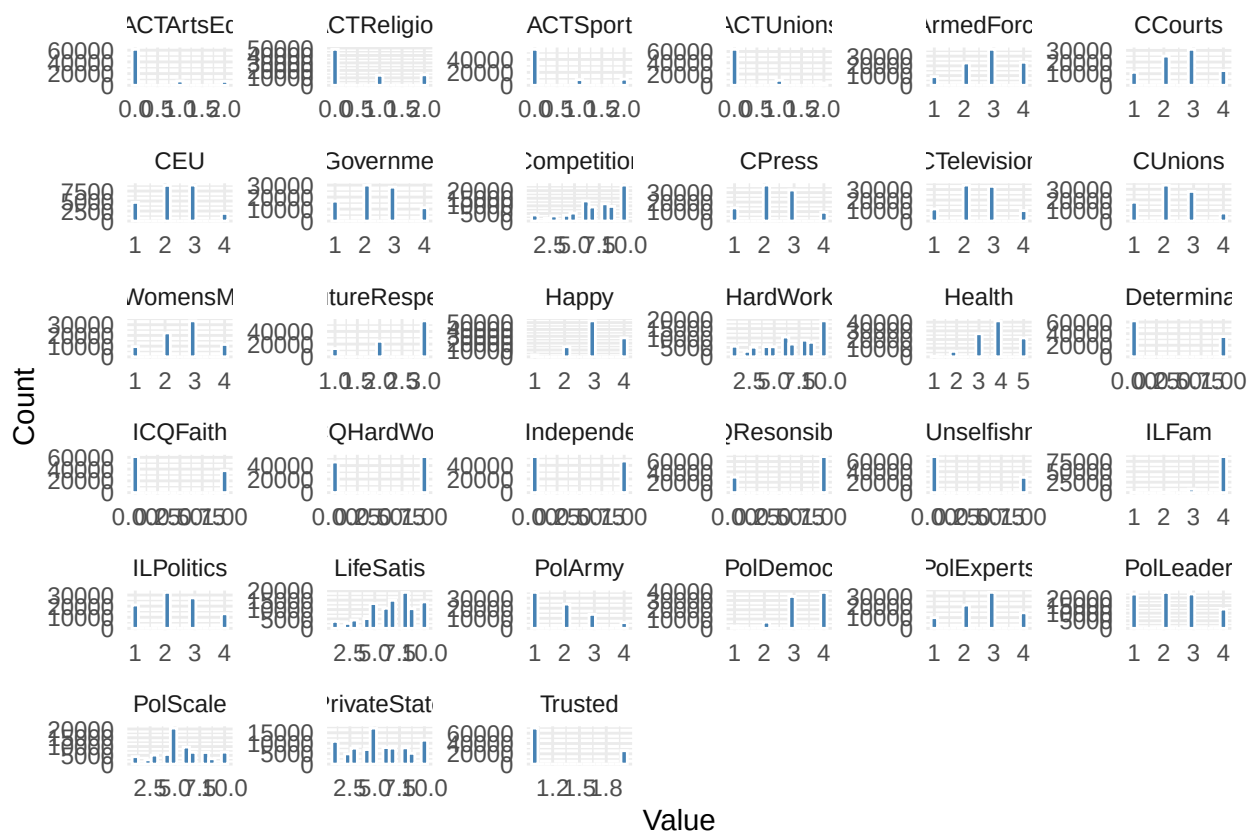
```
sort(sapply(VC_clean[sapply(VC_clean, is.numeric)], sd, na.rm = TRUE), decreasing = TRUE)
```

##	Year	HardWork	PrivateState	Competition
##	9.9252009	2.8798486	2.8504478	2.5839089
##	LifeSatis	PolScale	Wave	PolLeader
##	2.3907102	2.3692478	1.7066399	1.0385051
##	ILPolitics	PolArmy	CGovernment	CArmedForces
##	0.9764680	0.9456016	0.9402626	0.9240871
##	CCourts	PolExperts	Health	CUnions
##	0.9181461	0.9158981	0.8870618	0.8870584
##	CWomensMvt	CEU	CPress	CTelevision
##	0.8857883	0.8712353	0.8564255	0.8557747
##	ACTReligion	PolDemoc	Happy	FutureRespect
##	0.7875493	0.7512568	0.7366804	0.7246025
##	ACTSport	ACTArtsEd	ACTUnions	ICQIndependence
##	0.7016227	0.6140746	0.5228742	0.4989369
##	ICQHardWork	ICQFaith	ICQDetermination	ICQResonsibility
##	0.4983572	0.4849820	0.4800307	0.4582731
##	ICQUnselfishness	Trusted	ILFam	
##	0.4572204	0.4447817	0.3626752	

```
# Histograms for all numeric variables (excluding Wave/Year)
num_data <- VC_clean[, sapply(VC_clean, is.numeric)]
num_data <- num_data[, !names(num_data) %in% c("Wave", "Year")]
long <- pivot_longer(num_data, cols = everything(), names_to = "Variable", values_to = "Value")

ggplot(long, aes(x = Value)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 15) +
  facet_wrap(~Variable, scales = "free") +
  labs(x = "Value", y = "Count") +
  theme_minimal()
```

```
## Warning: Removed 460170 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Question 2(a)

```
# Q2(a) - SGP vs Others: Differences and Effect Sizes

# Separate SGP and non-SGP
sgp_data <- VC_clean[VC_clean$Country == "SGP", ]
oth_data <- VC_clean[VC_clean$Country != "SGP", ]
cat("SGP observations:", nrow(sgp_data), "\n")
```

```
## SGP observations: 1618
```

```

cat("Other observations:", nrow(oth_data), "\n")

## Other observations: 98382

# Variable groups
importance_in_life_vars <- c("ILFam", "ILPolitics")
child_vars <- c("ICQIndependence", "ICQHardWork", "ICQResonsibility",
               "ICQDetermination", "ICQFaith", "ICQUnselfishness")
membership_vars <- c("ACTReligion", "ACTSport", "ACTArtsEd", "ACTUnions")
wellbeing_vars <- c("Happy", "Health", "Trusted", "LifeSatis", "FutureRespect")
economic_vars <- c("PrivateState", "Competition", "HardWork")
political_system_vars <- c("PolLeader", "PolExperts", "PolArmy", "PolDemoc", "PolScale")

all_vars <- c(importance_in_life_vars, child_vars, membership_vars,
              wellbeing_vars, economic_vars, political_system_vars, confidence_vars)
conf_cols <- confidence_vars # used later for plot colouring

# Cohen's d
cohen_d <- function(x, y) {
  x <- na.omit(x); y <- na.omit(y)
  nx <- length(x); ny <- length(y)
  if (nx < 2 || ny < 2) return(NA_real_)
  pooled_sd <- sqrt(((nx - 1) * var(x) + (ny - 1) * var(y)) / (nx + ny - 2))
  if (pooled_sd == 0) return(0)
  (mean(x) - mean(y)) / pooled_sd
}

# Welch two-sample t-test p-value
welch_p <- function(x, y) {
  x <- na.omit(x); y <- na.omit(y)
  if (length(x) < 2 || length(y) < 2) return(NA_real_)
  tryCatch(t.test(x, y)$p.value, error = function(e) NA_real_)
}

# Build summary table
diff_tbl <- data.frame(
  Variable = all_vars,
  Type = ifelse(all_vars %in% conf_cols, "Confidence", "Attribute"),
  SGP_N = sapply(all_vars, function(v) sum(!is.na(sgp_data[[v]]))),
  Others_N = sapply(all_vars, function(v) sum(!is.na(oth_data[[v]]))),
  SGP_Mean = round(sapply(all_vars, function(v) mean(sgp_data[[v]], na.rm = TRUE)), 3),
  Others_Mean = round(sapply(all_vars, function(v) mean(oth_data[[v]], na.rm = TRUE)), 3),
  Cohen_d = round(sapply(all_vars, function(v) cohen_d(sgp_data[[v]], oth_data[[v]])), 3),
  p_value = sapply(all_vars, function(v) welch_p(sgp_data[[v]], oth_data[[v]])),
  row.names = NULL
)

diff_tbl$p_adj <- p.adjust(diff_tbl$p_value, method = "BH")
diff_tbl <- diff_tbl[order(-abs(diff_tbl$Cohen_d)), ]

diff_tbl$p_value <- ifelse(
  diff_tbl$p_value < 0.001,

```



```

"<0.001",
  round(diff_tbl$p_value, 3)
)

diff_tbl$p_adj<- ifelse(
  diff_tbl$p_adj < 0.001,
  "<0.001",
  round(diff_tbl$p_adj, 3)
)

rownames(diff_tbl) <- NULL

kable(diff_tbl, digits = 3,
      caption = "SGP vs Others: differences and effect sizes") |>
  kable_styling(bootstrap_options = c("striped", "hover"),
    full_width = FALSE)

```

Table 1: SGP vs Others: differences and effect sizes

Variable	Type	SGP_N	Others_N	SGP_Mean	Others_Mean	Cohen_d	p_value	p_adj
CGovernment	Confidence	1159	84905	3.004	2.385	0.660	<0.001	<0.001
CCourts	Confidence	1151	78795	3.017	2.556	0.503	<0.001	<0.001
ICQIndependence	Attribute	1618	97594	0.667	0.464	0.408	<0.001	<0.001
CUnions	Confidence	1122	86287	2.585	2.286	0.337	<0.001	<0.001
PolScale	Attribute	522	70173	4.944	5.700	-0.319	<0.001	<0.001
CPress	Confidence	1156	92649	2.627	2.396	0.270	<0.001	<0.001
Health	Attribute	1167	95910	4.001	3.794	0.233	<0.001	<0.001
Happy	Attribute	1617	96245	3.244	3.072	0.232	<0.001	<0.001
CArmedForces	Confidence	1159	91466	3.019	2.814	0.222	<0.001	<0.001
PolExperts	Attribute	1531	77932	2.462	2.664	-0.220	<0.001	<0.001
CTelevision	Confidence	1159	87601	2.638	2.464	0.203	<0.001	<0.001
ICQFaith	Attribute	1618	96977	0.294	0.380	-0.178	<0.001	<0.001
FutureRespect	Attribute	1603	92524	2.340	2.465	-0.173	<0.001	<0.001
CWomensMvt	Confidence	1051	76044	2.754	2.610	0.162	<0.001	<0.001
ILPolitics	Attribute	1615	92371	2.482	2.346	0.139	<0.001	<0.001
ICQDetermination	Attribute	1618	95342	0.425	0.359	0.138	<0.001	<0.001
PrivateState	Attribute	1610	86058	5.146	5.506	-0.126	<0.001	<0.001
ACTUnions	Attribute	1166	75421	0.158	0.219	-0.117	<0.001	<0.001
LifeSatis	Attribute	1615	96939	7.035	6.756	0.117	<0.001	<0.001
ACTReligion	Attribute	1167	74542	0.636	0.549	0.111	<0.001	<0.001
ILFam	Attribute	1617	94067	3.922	3.888	0.094	<0.001	<0.001
Trusted	Attribute	1608	94283	1.312	1.271	0.091	<0.001	<0.001
ICQResonsibility	Attribute	1618	97692	0.740	0.699	0.090	<0.001	<0.001
PolLeader	Attribute	1543	79943	2.221	2.313	-0.089	<0.001	<0.001
ICQHardWork	Attribute	1618	96978	0.582	0.540	0.084	<0.001	<0.001
PolArmy	Attribute	1573	77854	1.790	1.870	-0.084	<0.001	<0.001
PolDemoc	Attribute	1566	80634	3.271	3.330	-0.079	<0.001	<0.001
HardWork	Attribute	1165	76225	6.518	6.684	-0.058	0.019	0.021
Competition	Attribute	1612	83735	7.107	7.235	-0.050	0.017	0.019
ICQUnselfishness	Attribute	1618	96556	0.278	0.298	-0.045	0.069	0.073

ACTSport	Attribute	1166	72822	0.355	0.377	-0.031	0.269	0.278
ACTArtsEd	Attribute	1167	74465	0.276	0.275	0.001	0.966	0.966
CEU	Confidence	0	24506	NaN	2.333	NA	NA	NA

```
gt_table <- diff_tbl |>
  gt() |>
  tab_header(
    title = "SGP vs Others: differences and effect sizes"
  ) |>
  fmt_number(
    columns = c(SGP_Mean, Others_Mean, Cohen_d),
    decimals = 3
  )

print(gt_table)
```

```
## <div id="fiicjbvhir" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;
##   <style>#fiicjbvhir table {
##     font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Se
##   -webkit-font-smoothing: antialiased;
##   -moz-osx-font-smoothing: grayscale;
## }
##
## #fiicjbvhir thead, #fiicjbvhir tbody, #fiicjbvhir tfoot, #fiicjbvhir tr, #fiicjbvhir td, #fiicjbvhir
##   border-style: none;
## }
##
## #fiicjbvhir p {
##   margin: 0;
##   padding: 0;
## }
##
## #fiicjbvhir .gt_table {
##   display: table;
##   border-collapse: collapse;
##   line-height: normal;
##   margin-left: auto;
##   margin-right: auto;
##   color: #333333;
##   font-size: 16px;
##   font-weight: normal;
##   font-style: normal;
##   background-color: #FFFFFF;
##   width: auto;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #A8A8A8;
##   border-right-style: none;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #A8A8A8;
```

```

## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_caption {
## padding-top: 4px;
## padding-bottom: 4px;
## }
##
## #fiicjbvhir .gt_title {
## color: #333333;
## font-size: 125%;
## font-weight: initial;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## border-bottom-color: #FFFFFF;
## border-bottom-width: 0;
## }
##
## #fiicjbvhir .gt_subtitle {
## color: #333333;
## font-size: 85%;
## font-weight: initial;
## padding-top: 3px;
## padding-bottom: 5px;
## padding-left: 5px;
## padding-right: 5px;
## border-top-color: #FFFFFF;
## border-top-width: 0;
## }
##
## #fiicjbvhir .gt_heading {
## background-color: #FFFFFF;
## text-align: center;
## border-bottom-color: #FFFFFF;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_bottom_border {
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_col_headings {
## border-top-style: solid;

```

```

## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_col_heading {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 6px;
## padding-left: 5px;
## padding-right: 5px;
## overflow-x: hidden;
## }
##
## #fiicjbvhir .gt_column_spanner_outer {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## padding-top: 0;
## padding-bottom: 0;
## padding-left: 4px;
## padding-right: 4px;
## }
##
## #fiicjbvhir .gt_column_spanner_outer:first-child {
## padding-left: 0;
## }
##
## #fiicjbvhir .gt_column_spanner_outer:last-child {
## padding-right: 0;
## }
##
## #fiicjbvhir .gt_column_spanner {

```

```

## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 5px;
## overflow-x: hidden;
## display: inline-block;
## width: 100%;
## }
##
## #fiicjbvhir .gt_spanner_row {
## border-bottom-style: hidden;
## }
##
## #fiicjbvhir .gt_group_heading {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: initial;
## text-transform: inherit;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: middle;
## text-align: left;
## }
##
## #fiicjbvhir .gt_empty_group_heading {
## padding: 0.5px;
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: initial;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: middle;

```

```

## }
##
## #fiicjbvhir .gt_from_md > :first-child {
##   margin-top: 0;
## }
##
## #fiicjbvhir .gt_from_md > :last-child {
##   margin-bottom: 0;
## }
##
## #fiicjbvhir .gt_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   margin: 10px;
##   border-top-style: solid;
##   border-top-width: 1px;
##   border-top-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
##   vertical-align: middle;
##   overflow-x: hidden;
## }
##
## #fiicjbvhir .gt_stub {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #fiicjbvhir .gt_stub_row_group {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
##   vertical-align: top;

```

```

## }
##
## #fiicjbvhir .gt_row_group_first td {
##   border-top-width: 2px;
## }
##
## #fiicjbvhir .gt_row_group_first th {
##   border-top-width: 2px;
## }
##
## #fiicjbvhir .gt_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #fiicjbvhir .gt_first_summary_row {
##   border-top-style: solid;
##   border-top-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_first_summary_row.thick {
##   border-top-width: 2px;
## }
##
## #fiicjbvhir .gt_last_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_grand_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #fiicjbvhir .gt_first_grand_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;

```

```

## border-top-style: double;
## border-top-width: 6px;
## border-top-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_last_grand_summary_row_top {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## border-bottom-style: double;
## border-bottom-width: 6px;
## border-bottom-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_stripped {
## background-color: rgba(128, 128, 128, 0.05);
## }
##
## #fiicjbvhir .gt_table_body {
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_footnotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_footnote {
## margin: 0px;
## font-size: 90%;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## }
##
## #fiicjbvhir .gt_sourcenotes {
## color: #333333;
## background-color: #FFFFFF;

```



```

## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## }
##
## #fiicjbvhir .gt_sourcenote {
## font-size: 90%;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## }
##
## #fiicjbvhir .gt_left {
## text-align: left;
## }
##
## #fiicjbvhir .gt_center {
## text-align: center;
## }
##
## #fiicjbvhir .gt_right {
## text-align: right;
## font-variant-numeric: tabular-nums;
## }
##
## #fiicjbvhir .gt_font_normal {
## font-weight: normal;
## }
##
## #fiicjbvhir .gt_font_bold {
## font-weight: bold;
## }
##
## #fiicjbvhir .gt_font_italic {
## font-style: italic;
## }
##
## #fiicjbvhir .gt_super {
## font-size: 65%;
## }
##
## #fiicjbvhir .gt_footnote_marks {
## font-size: 75%;
## vertical-align: 0.4em;
## position: initial;
## }
##

```

```

## #fiicjbvhir .gt_asterisk {
##   font-size: 100%;
##   vertical-align: 0;
## }
##
## #fiicjbvhir .gt_indent_1 {
##   text-indent: 5px;
## }
##
## #fiicjbvhir .gt_indent_2 {
##   text-indent: 10px;
## }
##
## #fiicjbvhir .gt_indent_3 {
##   text-indent: 15px;
## }
##
## #fiicjbvhir .gt_indent_4 {
##   text-indent: 20px;
## }
##
## #fiicjbvhir .gt_indent_5 {
##   text-indent: 25px;
## }
##
## #fiicjbvhir .katex-display {
##   display: inline-flex !important;
##   margin-bottom: 0.75em !important;
## }
##
## #fiicjbvhir div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-g
##   height: 0px !important;
## }
## </style>
## <table class="gt_table" data-quarto-disable-processing="false" data-quarto-bootstrap="false">
##   <thead>
##     <tr class="gt_heading">
##       <td colspan="9" class="gt_heading gt_title gt_font_normal gt_bottom_border" style>SGP vs Other:
##     </tr>
##
##     <tr class="gt_col_headings">
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##     </tr>
##   </thead>
##   <tbody class="gt_table_body">
##     <tr><td headers="Variable" class="gt_row gt_left">CGovernment</td>

```

```

## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1159</td>
## <td headers="Others_N" class="gt_row gt_right">84905</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.004</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.385</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.660</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CCourts</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1151</td>
## <td headers="Others_N" class="gt_row gt_right">78795</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.017</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.556</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.503</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQIndependence</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">97594</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.667</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.464</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.408</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CUnions</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1122</td>
## <td headers="Others_N" class="gt_row gt_right">86287</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.585</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.286</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.337</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PolScale</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">522</td>
## <td headers="Others_N" class="gt_row gt_right">70173</td>
## <td headers="SGP_Mean" class="gt_row gt_right">4.944</td>
## <td headers="Others_Mean" class="gt_row gt_right">5.700</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.319</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CPress</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1156</td>
## <td headers="Others_N" class="gt_row gt_right">92649</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.627</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.396</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.270</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">Health</td>

```

```

## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1167</td>
## <td headers="Others_N" class="gt_row gt_right">95910</td>
## <td headers="SGP_Mean" class="gt_row gt_right">4.001</td>
## <td headers="Others_Mean" class="gt_row gt_right">3.794</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.233</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">Happy</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1617</td>
## <td headers="Others_N" class="gt_row gt_right">96245</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.244</td>
## <td headers="Others_Mean" class="gt_row gt_right">3.072</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.232</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CARmedForces</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1159</td>
## <td headers="Others_N" class="gt_row gt_right">91466</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.019</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.814</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.222</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PolExperts</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1531</td>
## <td headers="Others_N" class="gt_row gt_right">77932</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.462</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.664</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.220</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CTelevision</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1159</td>
## <td headers="Others_N" class="gt_row gt_right">87601</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.638</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.464</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.203</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQFaith</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">96977</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.294</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.380</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.178</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">FutureRespect</td>

```

```

## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1603</td>
## <td headers="Others_N" class="gt_row gt_right">92524</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.340</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.465</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.173</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">CWomensMvt</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">1051</td>
## <td headers="Others_N" class="gt_row gt_right">76044</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.754</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.610</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.162</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ILPolitics</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1615</td>
## <td headers="Others_N" class="gt_row gt_right">92371</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.482</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.346</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.139</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQDetermination</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">95342</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.425</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.359</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.138</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PrivateState</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1610</td>
## <td headers="Others_N" class="gt_row gt_right">86058</td>
## <td headers="SGP_Mean" class="gt_row gt_right">5.146</td>
## <td headers="Others_Mean" class="gt_row gt_right">5.506</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.126</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ACTUnions</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1166</td>
## <td headers="Others_N" class="gt_row gt_right">75421</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.158</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.219</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.117</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">LifeSatis</td>

```

```

## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1615</td>
## <td headers="Others_N" class="gt_row gt_right">96939</td>
## <td headers="SGP_Mean" class="gt_row gt_right">7.035</td>
## <td headers="Others_Mean" class="gt_row gt_right">6.756</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.117</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ACTReligion</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1167</td>
## <td headers="Others_N" class="gt_row gt_right">74542</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.636</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.549</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.111</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ILFam</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1617</td>
## <td headers="Others_N" class="gt_row gt_right">94067</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.922</td>
## <td headers="Others_Mean" class="gt_row gt_right">3.888</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.094</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">Trusted</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1608</td>
## <td headers="Others_N" class="gt_row gt_right">94283</td>
## <td headers="SGP_Mean" class="gt_row gt_right">1.312</td>
## <td headers="Others_Mean" class="gt_row gt_right">1.271</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.091</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQResonsibility</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">97692</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.740</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.699</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.090</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PolLeader</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1543</td>
## <td headers="Others_N" class="gt_row gt_right">79943</td>
## <td headers="SGP_Mean" class="gt_row gt_right">2.221</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.313</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.089</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQHardWork</td>

```

```

## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">96978</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.582</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.540</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.084</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PolArmy</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1573</td>
## <td headers="Others_N" class="gt_row gt_right">77854</td>
## <td headers="SGP_Mean" class="gt_row gt_right">1.790</td>
## <td headers="Others_Mean" class="gt_row gt_right">1.870</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.084</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">PolDemoc</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1566</td>
## <td headers="Others_N" class="gt_row gt_right">80634</td>
## <td headers="SGP_Mean" class="gt_row gt_right">3.271</td>
## <td headers="Others_Mean" class="gt_row gt_right">3.330</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.079</td>
## <td headers="p_value" class="gt_row gt_left">&lt;0.001</td>
## <td headers="p_adj" class="gt_row gt_left">&lt;0.001</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">HardWork</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1165</td>
## <td headers="Others_N" class="gt_row gt_right">76225</td>
## <td headers="SGP_Mean" class="gt_row gt_right">6.518</td>
## <td headers="Others_Mean" class="gt_row gt_right">6.684</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.058</td>
## <td headers="p_value" class="gt_row gt_left">0.019</td>
## <td headers="p_adj" class="gt_row gt_left">0.021</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">Competition</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1612</td>
## <td headers="Others_N" class="gt_row gt_right">83735</td>
## <td headers="SGP_Mean" class="gt_row gt_right">7.107</td>
## <td headers="Others_Mean" class="gt_row gt_right">7.235</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.050</td>
## <td headers="p_value" class="gt_row gt_left">0.017</td>
## <td headers="p_adj" class="gt_row gt_left">0.019</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ICQUnselfishness</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1618</td>
## <td headers="Others_N" class="gt_row gt_right">96556</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.278</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.298</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.045</td>
## <td headers="p_value" class="gt_row gt_left">0.069</td>
## <td headers="p_adj" class="gt_row gt_left">0.073</td></tr>
## <tr><td headers="Variable" class="gt_row gt_left">ACTSport</td>

```

```

## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1166</td>
## <td headers="Others_N" class="gt_row gt_right">72822</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.355</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.377</td>
## <td headers="Cohen_d" class="gt_row gt_right">-0.031</td>
## <td headers="p_value" class="gt_row gt_left">0.269</td>
## <td headers="p_adj" class="gt_row gt_left">0.278</td></tr>
##   <tr><td headers="Variable" class="gt_row gt_left">ACTArtsEd</td>
## <td headers="Type" class="gt_row gt_left">Attribute</td>
## <td headers="SGP_N" class="gt_row gt_right">1167</td>
## <td headers="Others_N" class="gt_row gt_right">74465</td>
## <td headers="SGP_Mean" class="gt_row gt_right">0.276</td>
## <td headers="Others_Mean" class="gt_row gt_right">0.275</td>
## <td headers="Cohen_d" class="gt_row gt_right">0.001</td>
## <td headers="p_value" class="gt_row gt_left">0.966</td>
## <td headers="p_adj" class="gt_row gt_left">0.966</td></tr>
##   <tr><td headers="Variable" class="gt_row gt_left">CEU</td>
## <td headers="Type" class="gt_row gt_left">Confidence</td>
## <td headers="SGP_N" class="gt_row gt_right">0</td>
## <td headers="Others_N" class="gt_row gt_right">24506</td>
## <td headers="SGP_Mean" class="gt_row gt_right">NaN</td>
## <td headers="Others_Mean" class="gt_row gt_right">2.333</td>
## <td headers="Cohen_d" class="gt_row gt_right">NA</td>
## <td headers="p_value" class="gt_row gt_left">NA</td>
## <td headers="p_adj" class="gt_row gt_left">NA</td></tr>
## </tbody>
##
## </table>
## </div>

```

```

gt(diff_tbl) |>
  tab_header(
    title = "SGP vs Others: Differences and Effect Sizes"
  ) |>
  tab_source_note(
    source_note = "Note: Variables sorted by Cohen's d.
p value and p adj rounded to 3 dp;
values <0.001 displayed as <0.001.
CEU excluded (0 SGP observations)."
  ) |>
  gt_save("SGP_vs_Others_effect_sizes.png")

```

file:///var/folders/ny/ckq7hnl57x5p8_sjp402mqh0000gn/T//RtmpmwSB1v/file3bd93bfa863a.html screenshot

Forest Plot: Cohen's d

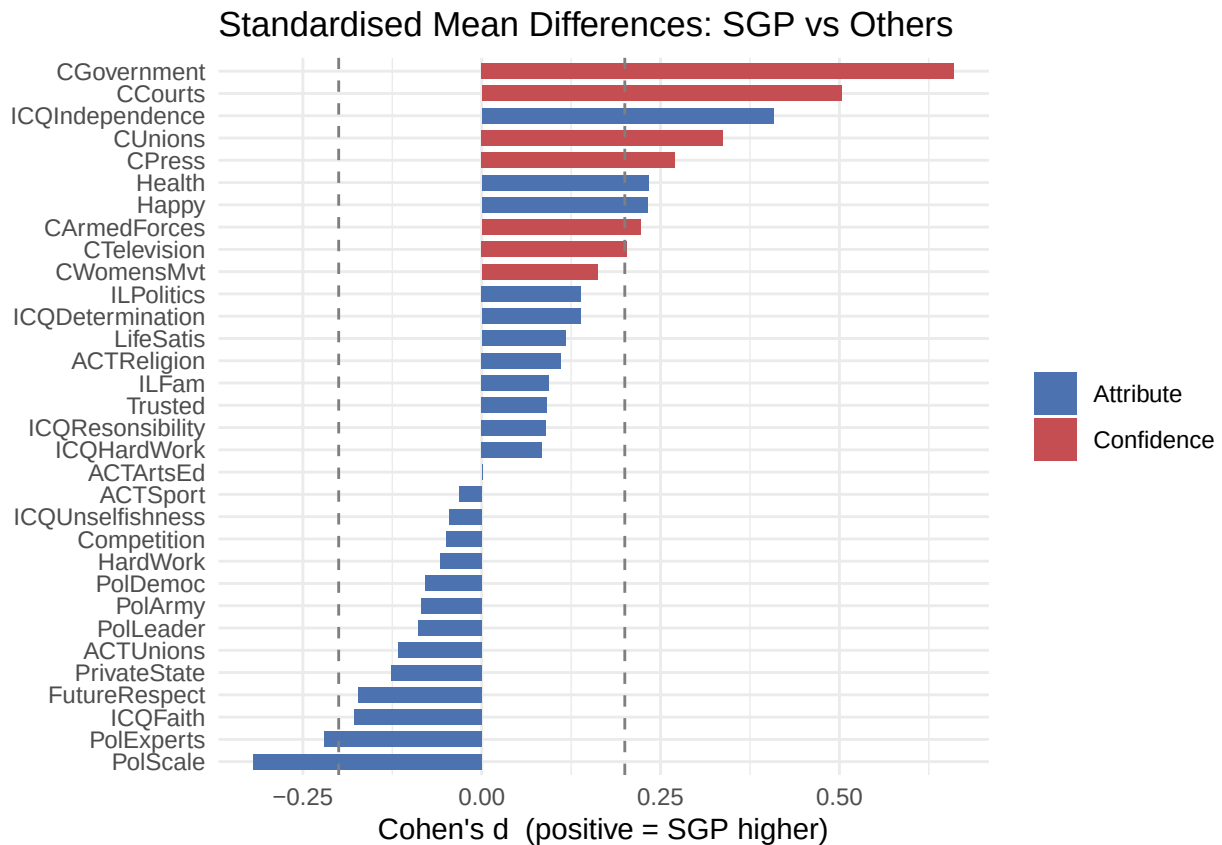
```

plot_dat <- diff_tbl[!is.na(diff_tbl$Cohen_d), ]
plot_dat$Variable <- factor(plot_dat$Variable,
  levels = plot_dat$Variable[order(plot_dat$Cohen_d)])
plot_dat$Type <- ifelse(plot_dat$Variable %in% conf_cols, "Confidence", "Attribute")

```



```
ggplot(plot_dat, aes(x = Cohen_d, y = Variable, fill = Type)) +
  geom_col(width = 0.7) +
  geom_vline(xintercept = c(-0.2, 0.2), linetype = "dashed", color = "grey50") +
  scale_fill_manual(values = c("Attribute" = "#4C72B0", "Confidence" = "#C44E52")) +
  labs(title = "Standardised Mean Differences: SGP vs Others",
       x = "Cohen's d (positive = SGP higher)",
       y = NULL,
       fill = NULL) +
  theme_minimal()
```



```
ggsave("standardised mean differences: sgp vs others.png",
       width = 10,
       height = 8,
       dpi = 300)
```

```
#Q2(b)
nrow(sgp_data)
```

```
## [1] 1618
```

```
x_vars <- setdiff(names(VC_clean),
                  c("Wave", "Country", "Year",
                    confidence_vars))
```

```
x_vars <- setdiff(x_vars, "is_SGP")
```

```
print(x_vars)
```

```
## [1] "ILFam"           "ILPolitics"      "ICQIndependence" "ICQHardWork"
## [5] "ICQResonsibility" "ICQDetermination" "ICQFaith"        "ICQUnselfishness"
## [9] "ACTReligion"      "ACTSport"        "ACTArtsEd"       "ACTUnions"
## [13] "Happy"           "Health"          "Trusted"         "LifeSatis"
## [17] "FutureRespect"    "PolScale"        "PrivateState"    "Competition"
## [21] "HardWork"        "PolLeader"       "PolExperts"      "PolArmy"
## [25] "PolDemoc"
```

```
length(x_vars)
```

```
## [1] 25
```

```
sgp_std <- sgp_data
```

```
num_to_scale <- c(x_vars, confidence_vars)
```

```
num_to_scale <- num_to_scale[sapply(sgp_std[num_to_scale], is.numeric)]
```

```
sgp_std[num_to_scale] <- scale(sgp_std[num_to_scale])
```

```
#multiple regression
```

```
#CGovernment
```

```
formula_gov <- as.formula(paste("CGovernment ~",  
                                paste(x_vars, collapse = " + ")))
```

```
# CArmedForces
```

```
formula_af <- as.formula(paste("CArmedForces ~", paste(x_vars, collapse = " + ")))
```

```
# CPress
```

```
formula_press <- as.formula(paste("CPress ~", paste(x_vars, collapse = " + ")))
```

```
# CUnions
```

```
formula_unions <- as.formula(paste("CUnions ~", paste(x_vars, collapse = " + ")))
```

```
# CTelevision
```

```
formula_tv <- as.formula(paste("CTelevision ~", paste(x_vars, collapse = " + ")))
```

```
# CWomensMvt
```

```
formula_wm <- as.formula(paste("CWomensMvt ~", paste(x_vars, collapse = " + ")))
```

```
# CCourts
```

```
formula_courts <- as.formula(paste("CCourts ~", paste(x_vars, collapse = " + ")))
```

```
# CEU
```

```
formula_eu <- as.formula(paste("CEU ~", paste(x_vars, collapse = " + ")))
```

```
model_gov <- lm(formula_gov, data = sgp_std)
```

```
model_af <- lm(formula_af, data = sgp_std)
```

```

model_press <- lm(formula_press, data = sgp_std)
model_unions <- lm(formula_unions, data = sgp_std)
model_tv <- lm(formula_tv, data = sgp_std)
model_wm <- lm(formula_wm, data = sgp_std)
model_courts <- lm(formula_courts, data = sgp_std)

models <- list(
  CGovernment = model_gov,
  CArmedForces = model_af,
  CPress = model_press,
  CUnions = model_unions,
  CTelevision = model_tv,
  CWomensMvt = model_wm,
  CCourts = model_courts
)

sgp_reg <- data.frame()

for (name in names(models)) {
  co <- summary(models[[name]])$coefficients
  co <- co[rownames(co) != "(Intercept)", , drop = FALSE]

  out <- data.frame(
    Confidence = name,
    Attribute = rownames(co),
    Beta       = NA_real_,
    t_value    = NA_real_,
    p          = NA_real_,
    row.names  = NULL
  )

  idx <- match(rownames(co), out$Attribute)

  out$Beta[idx] <- co[, "Estimate"]
  out$t_value[idx] <- co[, "t value"]
  out$p[idx] <- co[, "Pr(>|t|)"]

  sgp_reg <- rbind(sgp_reg, out)
}

# extract summary for every model
model_summary <- data.frame()

for (name in names(models)) {
  s <- summary(models[[name]])
  model_summary <- rbind(model_summary, data.frame(
    DV = name,
    R2 = round(s$r.squared, 3),
    Adj_R2 = round(s$adj.r.squared, 3),
    F_stat = round(s$fstatistic[1], 3),

```

```

    F_p_value = round(pf(s$fstatistic[1], s$fstatistic[2], s$fstatistic[3], lower.tail = FALSE), 4),
    Residual_Median = round(median(residuals(models[[name]])), 4)
  ))
}

sgp_reg <- sgp_reg[!is.na(sgp_reg$t_value), ]
sgp_reg <- sgp_reg[!is.na(sgp_reg$Beta), ]

# rank according to R2
model_summary <- model_summary[order(-model_summary$R2), ]

# display p-value
model_summary$F_p_value <- ifelse(model_summary$F_p_value < 0.001, "< 0.001", model_summary$F_p_value)

kable(model_summary,
  col.names = c("Dependent Variable", "R2", "Adj. R2", "F-statistic", "p-value", "Residual Median"),
  caption = "Model Summary: Multiple Regression for Confidence Variables (SGP Only)",
  align = c("l", "r", "r", "r", "r", "r"),
  row.names = FALSE) |>
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE)

```

Table 2: Model Summary: Multiple Regression for Confidence Variables (SGP Only)

Dependent Variable	R ²	Adj. R ²	F-statistic	p-value	Residual Median
CGovernment	0.226	0.183	5.181	< 0.001	0.0488
CTelevision	0.181	0.135	3.936	< 0.001	0.1112
CCourts	0.161	0.113	3.384	< 0.001	0.0192
CArmedForces	0.156	0.108	3.281	< 0.001	-0.0204
CPress	0.153	0.105	3.198	< 0.001	0.1432
CUnions	0.135	0.084	2.634	< 0.001	0.1582
CWomensMvt	0.093	0.032	1.529	0.0516	0.1834

```

sgp_reg <- data.frame()

for (name in names(models)) {

  s <- summary(models[[name]])
  coefs <- s$coefficients

  # skip if no predictors
  if (nrow(coefs) <= 1) next

  coefs <- coefs[-1, , drop = FALSE]

  tmp <- data.frame(
    Confidence = name,
    Attribute = rownames(coefs),
    Beta = as.numeric(coefs[, 1]),
    t_value = as.numeric(coefs[, 3]),

```

```

    p = as.numeric(coefs[, 4]),
    row.names = NULL
  )

  sgp_reg <- rbind(sgp_reg, tmp)
}

sgp_reg <- sgp_reg[
  !is.na(sgp_reg$Beta) &
  !is.na(sgp_reg$t_value) &
  !is.na(sgp_reg$Attribute),
]

safe_mean <- function(x) {
  if (length(x) == 0 || all(is.na(x))) return(NA_real_)
  mean(abs(x), na.rm = TRUE)
}

safe_sum <- function(x) {
  if (length(x) == 0 || all(is.na(x))) return(0)
  sum(x < 0.05, na.rm = TRUE)
}

sgp_imp <- aggregate(Beta ~ Attribute,
  data = sgp_reg,
  FUN = function(x) mean(abs(x), na.rm = TRUE))

names(sgp_imp)[2] <- "Mean_Abs_Beta"

sgp_imp_t <- aggregate(t_value ~ Attribute,
  data = sgp_reg,
  FUN = safe_mean)

names(sgp_imp_t)[2] <- "Mean_Abs_t"

sgp_sig <- aggregate(p ~ Attribute,
  data = sgp_reg,
  FUN = safe_sum)

names(sgp_sig)[2] <- "N_Sig"

sgp_imp <- Reduce(function(x, y)
  merge(x, y, by = "Attribute", all = TRUE),
  list(sgp_imp, sgp_imp_t, sgp_sig)
)

sgp_imp$Mean_Abs_Beta <- as.numeric(sgp_imp$Mean_Abs_Beta)
sgp_imp$Mean_Abs_t <- as.numeric(sgp_imp$Mean_Abs_t)

```

```

sgp_imp$N_Sig      <- as.numeric(sgp_imp$N_Sig)

sgp_imp$N_Sig[is.na(sgp_imp$N_Sig)] <- 0

sgp_imp$Mean_Abs_Beta <- round(sgp_imp$Mean_Abs_Beta, 3)
sgp_imp$Mean_Abs_t    <- round(sgp_imp$Mean_Abs_t, 3)

kable(sgp_imp,
      digits = 3,
      caption = "SGP: Attribute Predictive Importance",
      col.names = c("Attribute", "Mean_abs_Beta", "Mean_abs_t", "N Sig (p<.05)")) |>
kable_styling(full_width = FALSE) |>
footnote(general = paste0(
  "Based on ", length(models),
  " standardised regressions. Mean |Beta| = effect size; ",
  "Mean |t| = precision; N Sig counts p < 0.05 occurrences."
))

```

Table 3: SGP: Attribute Predictive Importance

Attribute	Mean_abs_Beta	Mean_abs_t	N Sig (p<.05)
ACTArtsEd	0.065	1.272	2
ACTReligion	0.031	0.617	0
ACTSport	0.025	0.446	0
ACTUnions	0.013	0.278	0
Competition	0.033	0.687	0
FutureRespect	0.072	1.861	2
Happy	0.119	2.200	4
HardWork	0.069	1.562	2
Health	0.030	0.614	0
ICQDetermination	0.084	1.853	3
ICQFaith	0.048	0.916	0
ICQHardWork	0.059	1.297	2
ICQIndependence	0.038	0.899	0
ICQResonsibility	0.060	1.403	2
ICQUnselfishness	0.048	1.136	1
ILFam	0.059	1.320	0
ILPolitics	0.062	1.351	1
LifeSatis	0.051	1.040	1
PolArmy	0.060	1.116	1
PolDemoc	0.111	2.669	5
PolExperts	0.089	1.844	3
PolLeader	0.035	0.699	0
PolScale	0.036	0.855	0
PrivateState	0.049	1.025	1

Trusted

0.093

2.264

5

Note:

Based on 7 standardised regressions. Mean $|\text{Beta}|$ = effect size; Mean $|t|$ = precision; N Sig counts $p < 0.05$ occurrences.

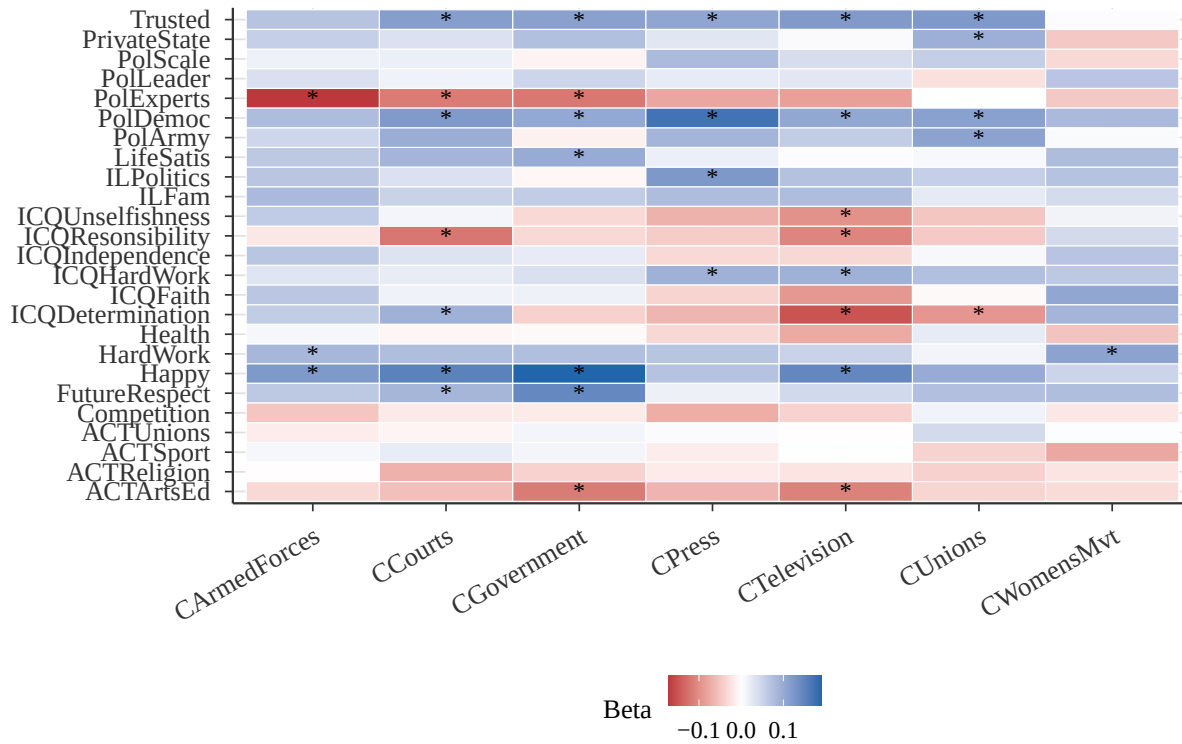
```
table_gt <- gt(sgp_imp) |>
  tab_header(title = "SGP: Attribute Predictive Importance")
gtsave(table_gt, "sgp_imp.png")
```

```
## file:///var/folders/ny/ckq7hnl57x5p8_sjp402mqh0000gn/T//RtmpmwSB1v/file3bd9635a8f96.html screenshot
```

```
sgp_reg$Sig <- ifelse(!is.na(sgp_reg$p) & sgp_reg$p < 0.05, "*", "")

ggplot(sgp_reg, aes(x = Confidence, y = Attribute, fill = Beta)) +
  geom_tile(color = "white", linewidth = 0.2) +
  geom_text(aes(label = Sig), size = 3) +
  scale_fill_gradient2(
    low = "#B2182B",
    mid = "white",
    high = "#2166AC",
    midpoint = 0
  ) +
  labs(title = "SGP: Standardised Regression Coefficients",
       x = NULL, y = NULL, fill = "Beta") +
  theme_pub +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```

SGP: Standardised Regression Coefficients



```
ggsave("sgp_reg_heatmap.png",
  width = 10,
  height = 8,
  dpi = 300)
```

Question 2(c)

```
# Q2(c) - Other Countries

# standardise
other_std <- oth_data

num_to_scale <- c(x_vars, confidence_vars)
num_to_scale <- num_to_scale[sapply(other_std[num_to_scale], is.numeric)]

other_std[num_to_scale] <- scale(other_std[num_to_scale])

models_o <- list()

for (y in confidence_vars) {
  vars <- c(y, x_vars)
```



```

sub <- other_std[, vars]

sub <- sub[complete.cases(sub), ]

if (nrow(sub) < 50) next

models_o[[y]] <- lm(as.formula(paste(y, "~ .")), data = sub)
}

model_summary_o <- data.frame()

for (name in names(models_o)) {
  s <- summary(models_o[[name]])
  model_summary_o <- rbind(model_summary_o, data.frame(
    DV = name,
    R2 = round(s$r.squared, 3),
    Adj_R2 = round(s$adj.r.squared, 3),
    F_stat = round(s$fstatistic[1], 3),
    F_p_value = round(pf(s$fstatistic[1], s$fstatistic[2], s$fstatistic[3], lower.tail = FALSE), 4),
    Residual_Median = round(median(residuals(models_o[[name]])), 4)
  ))
}

# Model summary - other

model_summary_o <- model_summary_o[order(-model_summary_o$R2), ]

model_summary_o$F_p_value <- ifelse(
  model_summary_o$F_p_value < 0.001,
  "< 0.001",
  model_summary_o$F_p_value
)

kable(model_summary_o,
  col.names = c("Dependent Variable", "R2", "Adj. R2",
    "F-statistic", "p-value", "Residual Mean"),
  caption = "Model Summary: Multiple Regression for Confidence Variables (Other Countries)",
  align = c("l", "r", "r", "r", "r", "r"),
  row.names = FALSE) |>
kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
  full_width = FALSE)

```

Table 4: Model Summary: Multiple Regression for Confidence Variables (Other Countries)

Dependent Variable	R ²	Adj. R ²	F-statistic	p-value	Residual Mean
CArmedForces	0.084	0.084	145.222	< 0.001	0.1375
CGovernment	0.080	0.079	137.304	< 0.001	-0.1024
CUnions	0.077	0.077	130.901	< 0.001	-0.1142

CCourts	0.065	0.064	109.277	< 0.001	0.1503
CTelevision	0.049	0.048	82.945	< 0.001	-0.2338
CWomensMvt	0.046	0.045	74.491	< 0.001	0.2361
CPress	0.043	0.043	72.991	< 0.001	-0.2219
CEU	0.035	0.033	18.221	< 0.001	-0.1789

```
table_gt <- model_summary_o |>
  gt() |>
  tab_header(
    title = "Model Summary: Multiple Regression for Confidence Variables (Other Countries)"
  )

gtsave(table_gt, "model_summary_other.png")
```

file:///var/folders/ny/ckq7hnln57x5p8_sjp402mqh0000gn/T//RtmpmwSB1v/file3bd9184b0aa9.html screenshot

```
#-----sgp Vs others comparison graphic-----#
# extract R²
sgp_r2 <- data.frame(
  Institution = names(models),
  R2 = sapply(models, function(m) summary(m)$r.squared),
  Group = "SGP"
)

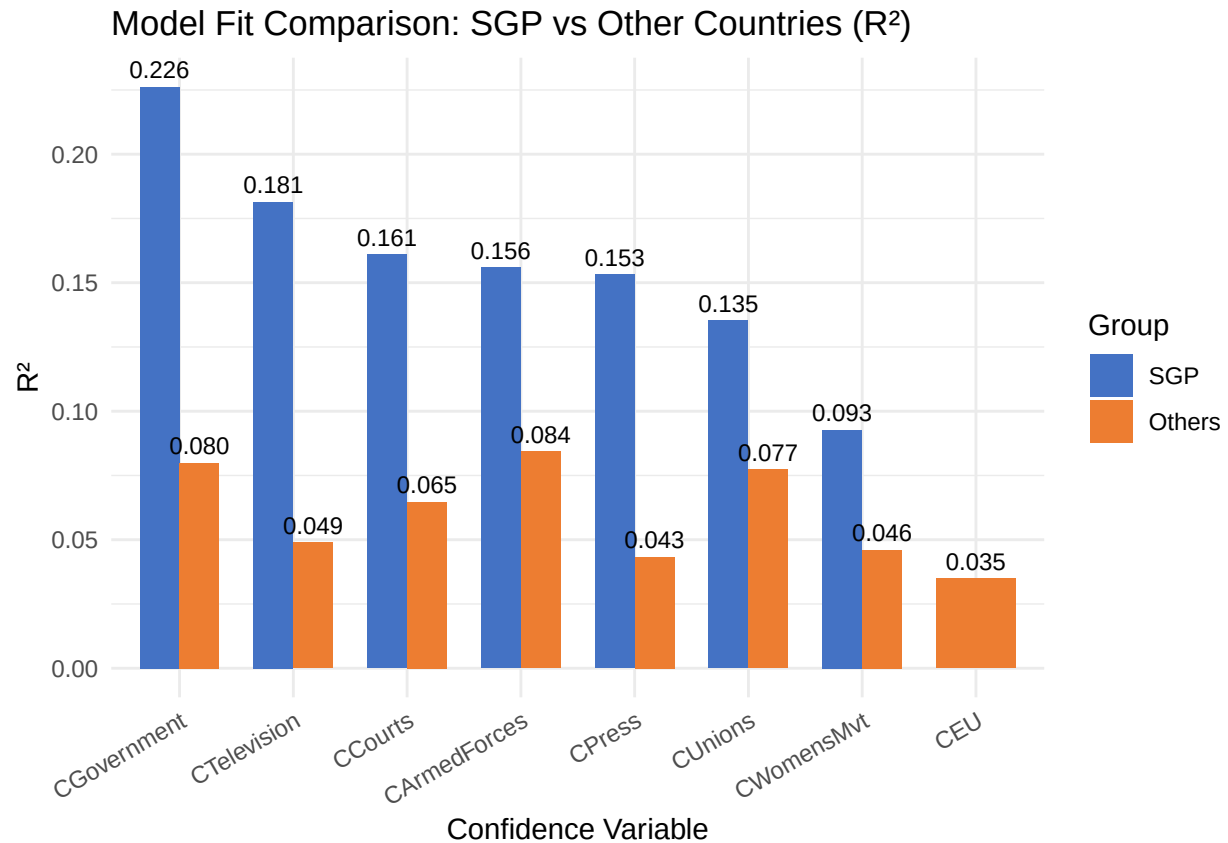
others_r2 <- data.frame(
  Institution = names(models_o),
  R2 = sapply(models_o, function(m) summary(m)$r.squared),
  Group = "Others"
)

r2_data <- rbind(sgp_r2, others_r2)

# order rank: SGP R² descending
sgp_order <- sgp_r2$Institution[order(-sgp_r2$R2)]
all_inst <- unique(c(sgp_order, setdiff(others_r2$Institution, sgp_r2$Institution)))
r2_data$Institution <- factor(r2_data$Institution, levels = all_inst)

# SGP on the left side factor level order control
r2_data$Group <- factor(r2_data$Group, levels = c("SGP", "Others"))

ggplot(r2_data, aes(x = Institution, y = R2, fill = Group)) +
  geom_col(position = "dodge", width = 0.7) +
  geom_text(aes(label = sprintf("%.3f", R2)),
    position = position_dodge(width = 0.7), vjust = -0.5, size = 3) +
  scale_fill_manual(values = c("SGP" = "#4472C4", "Others" = "#ED7D31")) +
  labs(title = "Model Fit Comparison: SGP vs Other Countries (R²)",
    x = "Confidence Variable", y = "R²", fill = "Group") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



```
reg_table_o <- data.frame()

for (name in names(models_o)) {
  s <- summary(models_o[[name]])
  coefs <- as.data.frame(s$coefficients)

  coefs <- coefs[-1, , drop = FALSE] # remove intercept

  reg_table_o <- rbind(
    reg_table_o,
    data.frame(
      Model = name,
      Predictor = rownames(coefs),
      Beta = coefs[, "Estimate"],
      p_value = coefs[, "Pr(>|t|)"],
      row.names = NULL
    )
  )
}

oth_reg <- reg_table_o[, c("Model", "Predictor", "Beta", "p_value")]

colnames(oth_reg) <- c("Confidence", "Attribute", "Beta", "p")

sgp_reg$Group <- "SGP"
oth_reg$Group <- "Others"
combo_reg <- rbind(sgp_reg[, c("Group", "Confidence", "Attribute", "Beta", "p")],
```

```

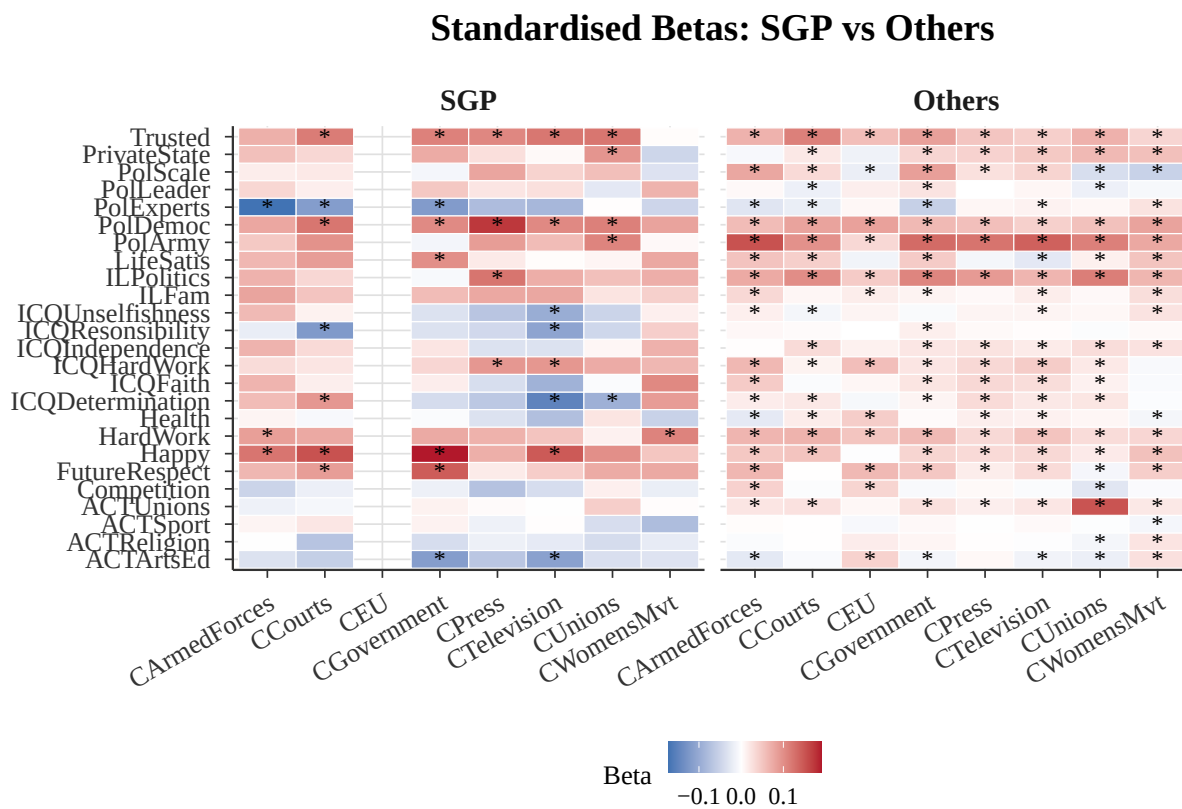
oth_reg[, c("Group", "Confidence", "Attribute", "Beta", "p")]
combo_reg$Group <- factor(combo_reg$Group, levels = c("SGP", "Others"))
combo_reg$Sig <- ifelse(!is.na(combo_reg$p) & combo_reg$p < 0.05, "*", "")

beta_comp_heatmap <- ggplot(combo_reg, aes(x = Confidence, y = Attribute, fill = Beta)) +
  geom_tile(color = "white", linewidth = 0.2) +
  geom_text(aes(label = Sig), size = 3) +
  facet_wrap(~ Group) +
  scale_fill_gradient2(low = "#2166AC", mid = "white", high = "#B2182B",
    midpoint = 0, na.value = "grey90") +
  labs(title = "Standardised Betas: SGP vs Others",
    x = NULL, y = NULL, fill = "Beta") +
  theme_pub +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))

ggsave("beta_heatmap_comparison.png", plot = beta_comp_heatmap, width = 10, height = 6, dpi = 300)

print(beta_comp_heatmap)

```



```

oth_reg_clean <- reg_table_o[
  !is.na(reg_table_o$Beta) &
  !is.na(reg_table_o$Predictor),
]

oth_imp <- aggregate(Beta ~ Predictor,

```

```

        data = oth_reg_clean,
        FUN = function(x) mean(abs(x), na.rm = TRUE))

names(oth_imp)[2] <- "Mean_Abs_Beta"

oth_imp_t <- aggregate(Beta ~ Predictor,
                      data = oth_reg_clean,
                      FUN = function(x) mean(abs(x), na.rm = TRUE))
names(oth_imp_t)[2] <- "Mean_Abs_t"

oth_sig <- aggregate(p_value ~ Predictor,
                    data = oth_reg_clean,
                    FUN = function(x) sum(x < 0.05, na.rm = TRUE))
names(oth_sig)[2] <- "N_Sig"

oth_imp <- Reduce(function(x, y)
  merge(x, y, by = "Predictor", all = TRUE),
  list(oth_imp, oth_imp_t, oth_sig)
)

names(oth_imp)[names(oth_imp) == "Predictor"] <- "Attribute"

other_imp <- oth_imp
str(other_imp)

## 'data.frame': 25 obs. of 4 variables:
## $ Attribute : chr "ACTArtsEd" "ACTReligion" "ACTSport" "ACTUnions" ...
## $ Mean_Abs_Beta: num 0.01845 0.00858 0.00466 0.0362 0.01651 ...
## $ Mean_Abs_t : num 0.01845 0.00858 0.00466 0.0362 0.01651 ...
## $ N_Sig : int 6 2 1 7 3 7 7 8 6 6 ...

```

```
nrow(other_imp)
```

```
## [1] 25
```

```
names(other_imp)
```

```
## [1] "Attribute" "Mean_Abs_Beta" "Mean_Abs_t" "N_Sig"
```

```

imp_cmp <- merge(
  data.frame(
    Attribute = sgp_imp$Attribute,
    SGP_Imp = sgp_imp$Mean_Abs_Beta,
    SGP_NSig = sgp_imp$N_Sig
  ),
  data.frame(
    Attribute = other_imp$Attribute,
    Other_Imp = other_imp$Mean_Abs_Beta,
    Other_NSig = other_imp$N_Sig
  ),
  by = "Attribute"

```

```

)
imp_cmp$Rank_SGP <- rank(-imp_cmp$SGP_Imp, ties.method = "first")
imp_cmp$Rank_Oth <- rank(-imp_cmp$Other_Imp, ties.method = "first")

imp_cmp <- imp_cmp[order(imp_cmp$Rank_SGP), ]
rownames(imp_cmp) <- NULL

#rank
rank_cor <- cor(imp_cmp$SGP_Imp, imp_cmp$Other_Imp,
               method = "spearman", use = "complete.obs")

kable(imp_cmp, digits = 3,
      caption = paste0("Attribute importance, Spearman rank r = ",
                      round(rank_cor, 3))) |>
  kable_styling(full_width = FALSE)

```

Table 5: Attribute importance, Spearman rank $r = 0.421$

Attribute	SGP_Imp	SGP_NSig	Other_Imp	Other_NSig	Rank_SGP	Rank_Oth
Happy	0.119	4	0.035	7	1	9
PolDemoc	0.111	5	0.065	8	2	4
Trusted	0.093	5	0.065	8	3	3
PolExperts	0.089	3	0.020	5	4	15
ICQDetermination	0.084	3	0.016	6	5	19
FutureRespect	0.072	2	0.035	7	6	10
HardWork	0.069	2	0.050	8	7	5
ACTArtsEd	0.065	2	0.018	6	8	16
ILPolitics	0.062	1	0.083	8	9	2
ICQResonsibility	0.060	2	0.005	1	10	24
PolArmy	0.060	1	0.109	8	11	1
ICQHardWork	0.059	2	0.032	7	12	11
ILFam	0.059	0	0.016	5	13	20
LifeSatis	0.051	1	0.032	6	14	12
PrivateState	0.049	1	0.036	6	15	8
ICQFaith	0.048	0	0.021	5	16	14
ICQUnselfishness	0.048	1	0.011	4	17	22
ICQIndependence	0.038	0	0.021	6	18	13
PolScale	0.036	0	0.046	8	19	6
PolLeader	0.035	0	0.012	3	20	21
Competition	0.033	0	0.017	3	21	18
ACTReligion	0.031	0	0.009	2	22	23
Health	0.030	0	0.017	6	23	17
ACTSport	0.025	0	0.005	1	24	25
ACTUnions	0.013	0	0.036	7	25	7

```

table_imp_comp <- imp_cmp |>
  gt() |>
  tab_header(title = "
Attribute importance, Spearman rank r = 0.417")

```

```
gtsave(table_imp_comp, "rank_att.png")
```

```
## file:///var/folders/ny/ckq7hnl57x5p8_sjp402mqh0000gn/T//RtmpmwSB1v/file3bd9178463d3.html screenshot
```

```
print(table_imp_comp)
```

```
## <div id="ydxqjogcps" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;
##   <style>#ydxqjogcps table {
##     font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Se
##     -webkit-font-smoothing: antialiased;
##     -moz-osx-font-smoothing: grayscale;
##   }
##
## #ydxqjogcps thead, #ydxqjogcps tbody, #ydxqjogcps tfoot, #ydxqjogcps tr, #ydxqjogcps td, #ydxqjogcps
##   border-style: none;
## }
##
## #ydxqjogcps p {
##   margin: 0;
##   padding: 0;
## }
##
## #ydxqjogcps .gt_table {
##   display: table;
##   border-collapse: collapse;
##   line-height: normal;
##   margin-left: auto;
##   margin-right: auto;
##   color: #333333;
##   font-size: 16px;
##   font-weight: normal;
##   font-style: normal;
##   background-color: #FFFFFF;
##   width: auto;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #A8A8A8;
##   border-right-style: none;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #A8A8A8;
##   border-left-style: none;
##   border-left-width: 2px;
##   border-left-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_caption {
##   padding-top: 4px;
##   padding-bottom: 4px;
## }
```

```

##
## #ydxqjogcps .gt_title {
##   color: #333333;
##   font-size: 125%;
##   font-weight: initial;
##   padding-top: 4px;
##   padding-bottom: 4px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-color: #FFFFFF;
##   border-bottom-width: 0;
## }
##
## #ydxqjogcps .gt_subtitle {
##   color: #333333;
##   font-size: 85%;
##   font-weight: initial;
##   padding-top: 3px;
##   padding-bottom: 5px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-top-color: #FFFFFF;
##   border-top-width: 0;
## }
##
## #ydxqjogcps .gt_heading {
##   background-color: #FFFFFF;
##   text-align: center;
##   border-bottom-color: #FFFFFF;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_bottom_border {
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_col_headings {
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;

```



```

## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_col_heading {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 6px;
## padding-left: 5px;
## padding-right: 5px;
## overflow-x: hidden;
## }
##
## #ydxqjogcps .gt_column_spanner_outer {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## padding-top: 0;
## padding-bottom: 0;
## padding-left: 4px;
## padding-right: 4px;
## }
##
## #ydxqjogcps .gt_column_spanner_outer:first-child {
## padding-left: 0;
## }
##
## #ydxqjogcps .gt_column_spanner_outer:last-child {
## padding-right: 0;
## }
##
## #ydxqjogcps .gt_column_spanner {
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 5px;
## overflow-x: hidden;
## display: inline-block;
## width: 100%;

```

```

## }
##
## #ydxqjogcps .gt_spanner_row {
##   border-bottom-style: hidden;
## }
##
## #ydxqjogcps .gt_group_heading {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
##   vertical-align: middle;
##   text-align: left;
## }
##
## #ydxqjogcps .gt_empty_group_heading {
##   padding: 0.5px;
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
##   vertical-align: middle;
## }
##
## #ydxqjogcps .gt_from_md > :first-child {
##   margin-top: 0;
## }
##
## #ydxqjogcps .gt_from_md > :last-child {
##   margin-bottom: 0;
## }

```

```

##
## #ydxqjogcps .gt_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   margin: 10px;
##   border-top-style: solid;
##   border-top-width: 1px;
##   border-top-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
##   vertical-align: middle;
##   overflow-x: hidden;
## }
##
## #ydxqjogcps .gt_stub {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #ydxqjogcps .gt_stub_row_group {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
##   vertical-align: top;
## }
##
## #ydxqjogcps .gt_row_group_first td {
##   border-top-width: 2px;
## }
##
## #ydxqjogcps .gt_row_group_first th {
##   border-top-width: 2px;
## }

```

```

##
## #ydxqjogcps .gt_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #ydxqjogcps .gt_first_summary_row {
##   border-top-style: solid;
##   border-top-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_first_summary_row.thick {
##   border-top-width: 2px;
## }
##
## #ydxqjogcps .gt_last_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_grand_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #ydxqjogcps .gt_first_grand_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-top-style: double;
##   border-top-width: 6px;
##   border-top-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_last_grand_summary_row_top {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;

```

```

## padding-right: 5px;
## border-bottom-style: double;
## border-bottom-width: 6px;
## border-bottom-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_stripped {
## background-color: rgba(128, 128, 128, 0.05);
## }
##
## #ydxqjogcps .gt_table_body {
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_footnotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## }
##
## #ydxqjogcps .gt_footnote {
## margin: 0px;
## font-size: 90%;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## }
##
## #ydxqjogcps .gt_sourcenotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;

```

```

## }
##
## #ydxqjogcps .gt_sourcenote {
##   font-size: 90%;
##   padding-top: 4px;
##   padding-bottom: 4px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #ydxqjogcps .gt_left {
##   text-align: left;
## }
##
## #ydxqjogcps .gt_center {
##   text-align: center;
## }
##
## #ydxqjogcps .gt_right {
##   text-align: right;
##   font-variant-numeric: tabular-nums;
## }
##
## #ydxqjogcps .gt_font_normal {
##   font-weight: normal;
## }
##
## #ydxqjogcps .gt_font_bold {
##   font-weight: bold;
## }
##
## #ydxqjogcps .gt_font_italic {
##   font-style: italic;
## }
##
## #ydxqjogcps .gt_super {
##   font-size: 65%;
## }
##
## #ydxqjogcps .gt_footnote_marks {
##   font-size: 75%;
##   vertical-align: 0.4em;
##   position: initial;
## }
##
## #ydxqjogcps .gt_asterisk {
##   font-size: 100%;
##   vertical-align: 0;
## }
##
## #ydxqjogcps .gt_indent_1 {
##   text-indent: 5px;
## }
##

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```

## #ydxqjogcps .gt_indent_2 {
##   text-indent: 10px;
## }
##
## #ydxqjogcps .gt_indent_3 {
##   text-indent: 15px;
## }
##
## #ydxqjogcps .gt_indent_4 {
##   text-indent: 20px;
## }
##
## #ydxqjogcps .gt_indent_5 {
##   text-indent: 25px;
## }
##
## #ydxqjogcps .katex-display {
##   display: inline-flex !important;
##   margin-bottom: 0.75em !important;
## }
##
## #ydxqjogcps div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-g
##   height: 0px !important;
## }
## </style>
## <table class="gt_table" data-quarto-disable-processing="false" data-quarto-bootstrap="false">
##   <thead>
##     <tr class="gt_heading">
##       <td colspan="7" class="gt_heading gt_title gt_font_normal gt_bottom_border" style>
## Attribute importance, Spearman rank r = 0.417</td>
##     </tr>
##
##     <tr class="gt_col_headings">
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##       <th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="co
##     </tr>
##   </thead>
##   <tbody class="gt_table_body">
##     <tr><td headers="Attribute" class="gt_row gt_left">Happy</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.119</td>
## <td headers="SGP_NSig" class="gt_row gt_right">4</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.034938125</td>
## <td headers="Other_NSig" class="gt_row gt_right">7</td>
## <td headers="Rank_SGP" class="gt_row gt_right">1</td>
## <td headers="Rank_Oth" class="gt_row gt_right">9</td></tr>
##     <tr><td headers="Attribute" class="gt_row gt_left">PolDemoc</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.111</td>
## <td headers="SGP_NSig" class="gt_row gt_right">5</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.065085362</td>

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## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">2</td>
## <td headers="Rank_Oth" class="gt_row gt_right">4</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">Trusted</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.093</td>
## <td headers="SGP_NSig" class="gt_row gt_right">5</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.065293808</td>
## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">3</td>
## <td headers="Rank_Oth" class="gt_row gt_right">3</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">PolExperts</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.089</td>
## <td headers="SGP_NSig" class="gt_row gt_right">3</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.020392775</td>
## <td headers="Other_NSig" class="gt_row gt_right">5</td>
## <td headers="Rank_SGP" class="gt_row gt_right">4</td>
## <td headers="Rank_Oth" class="gt_row gt_right">15</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ICQDetermination</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.084</td>
## <td headers="SGP_NSig" class="gt_row gt_right">3</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.016461024</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">5</td>
## <td headers="Rank_Oth" class="gt_row gt_right">19</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">FutureRespect</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.072</td>
## <td headers="SGP_NSig" class="gt_row gt_right">2</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.034588726</td>
## <td headers="Other_NSig" class="gt_row gt_right">7</td>
## <td headers="Rank_SGP" class="gt_row gt_right">6</td>
## <td headers="Rank_Oth" class="gt_row gt_right">10</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">HardWork</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.069</td>
## <td headers="SGP_NSig" class="gt_row gt_right">2</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.049838011</td>
## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">7</td>
## <td headers="Rank_Oth" class="gt_row gt_right">5</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ACTArtsEd</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.065</td>
## <td headers="SGP_NSig" class="gt_row gt_right">2</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.018451573</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">8</td>
## <td headers="Rank_Oth" class="gt_row gt_right">16</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ILPolitics</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.062</td>
## <td headers="SGP_NSig" class="gt_row gt_right">1</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.083201995</td>
## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">9</td>
## <td headers="Rank_Oth" class="gt_row gt_right">2</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ICQResonsibility</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.060</td>

```



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## <td headers="SGP_NSig" class="gt_row gt_right">2</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.005292520</td>
## <td headers="Other_NSig" class="gt_row gt_right">1</td>
## <td headers="Rank_SGP" class="gt_row gt_right">10</td>
## <td headers="Rank_Oth" class="gt_row gt_right">24</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">PolArmy</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.060</td>
## <td headers="SGP_NSig" class="gt_row gt_right">1</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.108689096</td>
## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">11</td>
## <td headers="Rank_Oth" class="gt_row gt_right">1</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ICQHardWork</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.059</td>
## <td headers="SGP_NSig" class="gt_row gt_right">2</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.032437241</td>
## <td headers="Other_NSig" class="gt_row gt_right">7</td>
## <td headers="Rank_SGP" class="gt_row gt_right">12</td>
## <td headers="Rank_Oth" class="gt_row gt_right">11</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ILFam</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.059</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.015668810</td>
## <td headers="Other_NSig" class="gt_row gt_right">5</td>
## <td headers="Rank_SGP" class="gt_row gt_right">13</td>
## <td headers="Rank_Oth" class="gt_row gt_right">20</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">LifeSatis</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.051</td>
## <td headers="SGP_NSig" class="gt_row gt_right">1</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.032158150</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">14</td>
## <td headers="Rank_Oth" class="gt_row gt_right">12</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">PrivateState</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.049</td>
## <td headers="SGP_NSig" class="gt_row gt_right">1</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.035621163</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">15</td>
## <td headers="Rank_Oth" class="gt_row gt_right">8</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ICQFaith</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.048</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.020500093</td>
## <td headers="Other_NSig" class="gt_row gt_right">5</td>
## <td headers="Rank_SGP" class="gt_row gt_right">16</td>
## <td headers="Rank_Oth" class="gt_row gt_right">14</td></tr>
## <tr><td headers="Attribute" class="gt_row gt_left">ICQUnselfishness</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.048</td>
## <td headers="SGP_NSig" class="gt_row gt_right">1</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.011277954</td>
## <td headers="Other_NSig" class="gt_row gt_right">4</td>
## <td headers="Rank_SGP" class="gt_row gt_right">17</td>
## <td headers="Rank_Oth" class="gt_row gt_right">22</td></tr>

```

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##      <tr><td headers="Attribute" class="gt_row gt_left">ICQIndependence</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.038</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.020718876</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">18</td>
## <td headers="Rank_Oth" class="gt_row gt_right">13</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">PolScale</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.036</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.046408163</td>
## <td headers="Other_NSig" class="gt_row gt_right">8</td>
## <td headers="Rank_SGP" class="gt_row gt_right">19</td>
## <td headers="Rank_Oth" class="gt_row gt_right">6</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">PolLeader</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.035</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.011974652</td>
## <td headers="Other_NSig" class="gt_row gt_right">3</td>
## <td headers="Rank_SGP" class="gt_row gt_right">20</td>
## <td headers="Rank_Oth" class="gt_row gt_right">21</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">Competition</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.033</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.016509880</td>
## <td headers="Other_NSig" class="gt_row gt_right">3</td>
## <td headers="Rank_SGP" class="gt_row gt_right">21</td>
## <td headers="Rank_Oth" class="gt_row gt_right">18</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">ACTReligion</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.031</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.008575757</td>
## <td headers="Other_NSig" class="gt_row gt_right">2</td>
## <td headers="Rank_SGP" class="gt_row gt_right">22</td>
## <td headers="Rank_Oth" class="gt_row gt_right">23</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">Health</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.030</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.017028425</td>
## <td headers="Other_NSig" class="gt_row gt_right">6</td>
## <td headers="Rank_SGP" class="gt_row gt_right">23</td>
## <td headers="Rank_Oth" class="gt_row gt_right">17</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">ACTSport</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.025</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.004660910</td>
## <td headers="Other_NSig" class="gt_row gt_right">1</td>
## <td headers="Rank_SGP" class="gt_row gt_right">24</td>
## <td headers="Rank_Oth" class="gt_row gt_right">25</td></tr>
##      <tr><td headers="Attribute" class="gt_row gt_left">ACTUnions</td>
## <td headers="SGP_Imp" class="gt_row gt_right">0.013</td>
## <td headers="SGP_NSig" class="gt_row gt_right">0</td>
## <td headers="Other_Imp" class="gt_row gt_right">0.036195058</td>
## <td headers="Other_NSig" class="gt_row gt_right">7</td>

```

```
## <td headers="Rank_SGP" class="gt_row gt_right">25</td>
## <td headers="Rank_0th" class="gt_row gt_right">7</td></tr>
## </tbody>
##
## </table>
## </div>
```

Question 3

```
# -----q3(a)
#determine using year/wave
#check if country has data for every year -
table(VC_clean$Country, VC_clean$Year)
```

```
##
##      1981 1982 1984 1989 1990 1991 1995 1996 1997 1998 1999 2000 2001 2002
## ARG      0      0 296      0      0 267 310      0      0      0 362      0      0      0
## ARM      0      0      0      0      0      0      0      0 538      0      0      0      0      0
## AUS 346      0      0      0      0      0      0 558      0      0      0      0      0      0
## BGD      0      0      0      0      0      0      0 431      0      0      0      0      0 417
## BLR      0      0      0      0 314      0      0 580      0      0      0      0      0      0
## BRA      0      0      0      0      0 527      0      0 332      0      0      0      0      0
## CAN      0      0      0      0      0      0      0      0      0      0      0 559      0      0
## CHE      0      0      0 394      0      0      0 324      0      0      0      0      0      0
## CHL      0      0      0      0 420      0      0 279      0      0      0 323      0      0
## CHN      0      0      0      0 280      0 431      0      0      0      0      0 278      0
## COL      0      0      0      0      0      0      0      0 843 843      0      0      0      0
## CYP      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## CZE      0      0      0      0      0 251      0      0      0 309      0      0      0      0
## DEU      0      0      0      0      0      0      0      0 575      0      0      0      0      0
## EGY      0      0      0      0      0      0      0      0      0      0      0      0 861      0
## ESP      0      0      0      0 420      0 337      0      0      0      0 354      0      0
## FIN 327      0      0      0      0      0      0      0 273      0      0      0      0      0
## GBR      0      0      0      0      0      0      0      0      0 334      0      0      0      0
## GEO      0      0      0      0      0      0      0 591      0      0      0      0      0      0
## HKG      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## HUN      0 414      0      0      0      0      0      0      0 173      0      0      0      0
## IDN      0      0      0      0      0      0      0      0      0      0      0      0 295      0
## IND      0      0      0      0 660      0 568      0      0      0      0      0 631      0
## IRN      0      0      0      0      0      0      0      0      0      0      0 707      0      0
## IRQ      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## JOR      0      0      0      0      0      0      0      0      0      0      0      0 339      0
## JPN 303      0      0      0 284      0 280      0      0      0      0 375      0      0
## KGZ      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## KOR      0 263      0      0 324      0      0 386      0      0      0      0 357      0
## MAR      0      0      0      0      0      0      0      0      0      0      0      0 336      0
## MDA      0      0      0      0      0      0      0 286      0      0      0      0      0 266
## MEX 475      0      0      0 437      0      0 427      0      0      0 387      0      0
## MYS      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## NGA      0      0      0      0 275      0 556      0      0      0      0 587      0      0
## NLD      0      0      0      0      0      0      0      0      0      0      0      0      0      0
## NZL      0      0      0      0      0      0      0      0      0 322      0      0      0      0
## PAK      0      0      0      0      0      0      0      0 199      0      0      0 562      0
```

##	PER	0	0	0	0	0	0	0	350	0	0	0	0	410	0
##	PHL	0	0	0	0	0	0	0	341	0	0	0	0	344	0
##	POL	0	0	0	269	0	0	0	0	318	0	0	0	0	0
##	PRI	0	0	0	0	0	0	338	0	0	0	0	0	208	0
##	ROU	0	0	0	0	0	0	0	0	0	329	0	0	0	0
##	RUS	0	0	0	0	567	0	568	0	0	0	0	0	0	0
##	SGP	0	0	0	0	0	0	0	0	0	0	0	0	0	450
##	SRB	0	0	0	0	0	0	0	365	0	0	0	0	320	0
##	SVK	0	0	0	0	132	0	0	0	0	332	0	0	0	0
##	SVN	0	0	0	0	0	0	277	0	0	0	0	0	0	0
##	SWE	0	0	0	0	0	0	0	301	0	0	270	0	0	0
##	THA	0	0	0	0	0	0	0	0	0	0	0	0	0	0
##	TUR	0	0	0	0	301	0	0	554	0	0	0	0	970	0
##	TWN	0	0	0	0	0	0	0	0	0	207	0	0	0	0
##	UKR	0	0	0	0	0	0	0	753	0	0	0	0	0	0
##	URY	0	0	0	0	0	0	0	289	0	0	0	0	0	0
##	USA	0	0	0	0	0	0	469	0	0	0	344	0	0	0
##	VEN	0	0	0	0	0	0	0	338	0	0	0	346	0	0
##	VNM	0	0	0	0	0	0	0	0	0	0	0	0	255	0
##	ZAF	0	432	0	0	788	0	0	780	0	0	0	0	827	0
##	ZWE	0	0	0	0	0	0	0	0	0	0	0	0	301	0
##															
##		2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2017	2018
##	ARG	0	0	0	281	0	0	0	0	0	0	286	0	294	0
##	ARM	0	0	0	0	0	0	0	0	310	0	0	0	0	0
##	AUS	0	0	423	0	0	0	0	0	0	406	0	0	0	525
##	BGD	0	0	0	0	0	0	0	0	0	0	0	0	0	339
##	BLR	0	0	0	0	0	0	0	0	416	0	0	0	0	0
##	BRA	0	0	0	432	0	0	0	0	0	0	0	424	0	499
##	CAN	0	0	0	621	0	0	0	0	0	0	0	0	0	0
##	CHE	0	0	0	0	350	0	0	0	0	0	0	0	0	0
##	CHL	0	0	0	301	0	0	0	0	0	270	0	0	0	273
##	CHN	0	0	0	0	570	0	0	0	0	0	610	0	0	856
##	COL	0	0	844	0	0	0	0	0	0	405	0	0	0	427
##	CYP	0	0	0	283	0	0	0	0	290	0	0	0	0	0
##	CZE	0	0	0	0	0	0	0	0	0	0	0	0	0	0
##	DEU	0	0	0	566	0	0	0	0	0	0	538	0	0	455
##	EGY	0	0	0	0	0	832	0	0	0	0	400	0	0	367
##	ESP	0	0	0	0	327	0	0	0	350	0	0	0	0	0
##	FIN	0	0	273	0	0	0	0	0	0	0	0	0	0	0
##	GBR	0	0	281	0	0	0	0	0	0	0	0	0	0	0
##	GEO	0	0	0	0	0	0	402	0	0	0	0	381	0	0
##	HKG	0	0	340	0	0	0	0	0	0	0	0	269	0	633
##	HUN	0	0	0	0	0	0	283	0	0	0	0	0	0	0
##	IDN	0	0	0	570	0	0	0	0	0	0	0	0	0	862
##	IND	0	0	0	583	0	0	0	0	0	1089	0	0	0	0
##	IRN	0	0	0	0	749	0	0	0	0	0	0	0	0	0
##	IRQ	0	699	0	728	0	0	0	0	0	0	332	0	0	348
##	JOR	0	0	0	0	327	0	0	0	0	0	0	358	0	348
##	JPN	0	0	293	0	0	0	0	677	0	0	0	0	0	0
##	KGZ	276	0	0	0	0	0	0	0	428	0	0	0	0	0
##	KOR	0	0	324	0	0	0	0	329	0	0	0	0	0	345
##	MAR	0	0	0	0	342	0	0	0	327	0	0	0	0	0
##	MDA	0	0	0	267	0	0	0	0	0	0	0	0	0	0

##	MEX	0	0	418	0	0	0	0	0	0	582	0	0	0	517
##	MYS	0	0	0	336	0	0	0	0	0	376	0	0	0	375
##	NGA	0	0	0	0	0	0	0	0	0	498	0	0	0	351
##	NLD	0	0	0	276	0	0	0	0	0	530	0	0	0	0
##	NZL	0	281	0	0	0	0	0	0	241	0	0	0	0	0
##	PAK	0	0	0	0	0	0	0	0	0	317	0	0	0	584
##	PER	0	0	0	425	0	0	0	0	0	338	0	0	0	367
##	PHL	0	0	0	0	0	0	0	0	0	340	0	0	0	0
##	POL	0	0	294	0	0	0	0	0	0	259	0	0	0	0
##	PRI	0	0	0	0	0	0	0	0	0	0	0	0	0	351
##	ROU	0	0	493	0	0	0	0	0	0	418	0	0	0	367
##	RUS	0	0	0	586	0	0	0	0	685	0	0	0	502	0
##	SGP	0	0	0	0	0	0	0	0	0	575	0	0	0	0
##	SRB	0	0	0	299	0	0	0	0	0	0	0	0	258	0
##	SVK	0	0	0	0	0	0	0	0	0	0	0	0	0	0
##	SVN	0	0	275	0	0	0	0	0	305	0	0	0	0	0
##	SWE	0	0	0	255	0	0	0	0	330	0	0	0	0	0
##	THA	0	0	0	0	427	0	0	0	0	0	341	0	0	428
##	TUR	0	0	0	0	383	0	0	0	443	0	0	0	0	639
##	TWN	0	0	0	341	0	0	0	0	0	341	0	0	0	0
##	UKR	0	0	0	266	0	0	0	0	421	0	0	0	0	0
##	URY	0	0	0	288	0	0	0	0	300	0	0	0	0	0
##	USA	0	0	0	346	0	0	0	0	640	0	0	0	723	0
##	VEN	0	0	0	0	0	0	0	0	0	0	0	0	0	0
##	VNM	0	0	0	373	0	0	0	0	0	0	0	0	0	0
##	ZAF	0	0	0	853	0	0	0	0	0	0	941	0	0	0
##	ZWE	0	0	0	0	0	0	0	0	0	391	0	0	0	0
##															
##		2019	2020	2021	2022	2023									
##	ARG	0	0	0	0	0									
##	ARM	0	0	353	0	0									
##	AUS	0	0	0	0	0									
##	BGD	0	0	0	0	0									
##	BLR	0	0	0	0	0									
##	BRA	0	0	0	0	0									
##	CAN	0	1065	0	0	0									
##	CHE	0	0	0	0	0									
##	CHL	0	0	0	0	0									
##	CHN	0	0	0	0	0									
##	COL	0	0	0	0	0									
##	CYP	278	0	0	0	0									
##	CZE	0	0	0	334	0									
##	DEU	0	0	0	0	0									
##	EGY	0	0	0	0	0									
##	ESP	0	0	0	0	0									
##	FIN	0	0	0	0	0									
##	GBR	0	0	0	744	0									
##	GEO	0	0	0	0	0									
##	HKG	0	0	0	0	0									
##	HUN	0	0	0	0	0									
##	IDN	0	0	0	0	0									
##	IND	0	0	0	0	476									
##	IRN	0	411	0	0	0									
##	IRQ	0	0	0	0	0									

```
## JOR 0 0 0 0 0
## JPN 375 0 0 0 0
## KGZ 0 360 0 0 0
## KOR 0 0 0 0 0
## MAR 0 0 320 0 0
## MDA 0 0 0 0 0
## MEX 0 0 0 0 0
## MYS 0 0 0 0 0
## NGA 0 0 0 0 0
## NLD 0 0 0 583 0
## NZL 0 319 0 0 0
## PAK 0 0 0 0 0
## PER 0 0 0 0 0
## PHL 350 0 0 0 0
## POL 0 0 0 0 0
## PRI 0 0 0 0 0
## ROU 0 0 0 0 0
## RUS 0 0 0 0 0
## SGP 0 593 0 0 0
## SRB 0 0 0 0 0
## SVK 0 0 0 338 0
## SVN 0 0 0 0 0
## SWE 0 0 0 0 0
## THA 0 0 0 0 0
## TUR 0 0 0 0 0
## TWN 353 0 0 0 0
## UKR 0 338 0 0 0
## URY 0 0 0 288 0
## USA 0 0 0 0 0
## VEN 0 0 344 0 0
## VNM 0 319 0 0 0
## ZAF 0 0 0 0 0
## ZWE 0 336 0 0 0
```

```
#-----spg wave mean
# common waves
common_waves <- c(4, 6, 7)
sgp_common <- sgp_data[sgp_data$Wave %in% common_waves, ]

wave_mean <- function(data, vars) {
  waves <- sort(unique(data$Wave))
  rows <- list()

  for (v in vars) {
    for (w in waves) {
      x <- data[[v]][data$Wave == w]
      x <- x[!is.na(x)]

      if (length(x) == 0) next

      rows[[paste(v, w, sep = "_")]] <- data.frame(
        Variable = v,
        Wave = w,
```

```

    Mean = mean(x),
    SD = sd(x),
    N = length(x),
    SE = ifelse(length(x) > 1, sd(x)/sqrt(length(x)), NA_real_)
  )
}
}

do.call(rbind, rows)
}

trend_test <- function(data, vars) {
  out <- data.frame(
    Variable = vars,
    slope = NA_real_,
    p = NA_real_,
    n_waves = NA_integer_
  )

  for (i in seq_along(vars)) {
    v <- vars[i]
    sub <- data[!is.na(data[[v]]), c("Wave", v)]

    out$n_waves[i] <- length(unique(sub$Wave))

    if (nrow(sub) < 30 || out$n_waves[i] < 2) next

    fit <- lm(sub[[v]] ~ sub$Wave)
    cf <- summary(fit)$coefficients

    if (nrow(cf) >= 2) {
      out$slope[i] <- cf[2,1]
      out$p[i] <- cf[2,4]
    }
  }

  out
}

sgp_wm <- wave_mean(sgp_data, all_vars)
sgp_wm$Group <- "SGP"

oth_wm <- wave_mean(oth_data, all_vars)
oth_wm$Group <- "Others"

wm_all <- rbind(sgp_wm, oth_wm)
wm_all$Group <- factor(wm_all$Group, levels = c("SGP", "Others"))

sgp_tr <- trend_test(sgp_data, all_vars)
oth_tr <- trend_test(oth_data, all_vars)

```

```

trend_wide <- merge(
  sgp_tr,
  oth_tr,
  by = "Variable",
  suffixes = c("_SGP", "_OTH")
)

trend_wide$Type <- ifelse(
  trend_wide$Variable %in% conf_cols,
  "Confidence",
  "Attribute"
)

trend_wide <- trend_wide[order(trend_wide$p_SGP), ]
rownames(trend_wide) <- NULL

trend_wide$p_SGP <- round(trend_wide$p_SGP, 3)
trend_wide$p_OTH <- round(trend_wide$p_OTH, 3)

trend_wide$p_SGP <- ifelse(trend_wide$p_SGP < 0.001, 0.001, "<0.001")
trend_wide$p_OTH <- ifelse(trend_wide$p_OTH < 0.001, 0.001, "<0.001")

kable(trend_wide, digits = 3,
      caption = "Wave trend coefficients: SGP vs Others") |>
  kable_styling(full_width = FALSE)

```

Table 6: Wave trend coefficients: SGP vs Others

Variable	slope_SGP	p_SGP	n_waves_SGP	slope_OTH	p_OTH	n_waves_OTH	Type
Trusted	0.077	0.001	3	-0.005	0.001	7	Attribute
ICQFaith	-0.068	0.001	3	0.001	<0.001	7	Attribute
FutureRespect	-0.082	0.001	3	-0.008	0.001	7	Attribute
Competition	-0.216	0.001	3	-0.140	0.001	6	Attribute
ACTSport	-0.185	0.001	2	-0.011	0.001	4	Attribute
Health	-0.204	0.001	2	0.029	0.001	7	Attribute
PolArmy	-0.077	0.001	3	0.057	0.001	5	Attribute
ICQIndependence	-0.042	0.001	3	0.023	0.001	7	Attribute
Happy	-0.047	0.001	3	0.032	0.001	7	Attribute
ICQResonsibility	-0.032	0.001	3	0.007	0.001	7	Attribute
ICQHardWork	-0.035	<0.001	3	0.012	0.001	7	Attribute
ACTArtsEd	-0.111	<0.001	2	0.007	0.001	5	Attribute
CUnions	-0.121	<0.001	2	0.004	<0.001	7	Confidence
LifeSatis	-0.103	<0.001	3	0.091	0.001	7	Attribute
PolLeader	0.046	<0.001	3	0.069	0.001	5	Attribute
PolExperts	0.038	<0.001	3	0.006	<0.001	5	Attribute
ACTUnions	-0.052	<0.001	2	-0.008	0.001	5	Attribute
CTelevision	-0.088	<0.001	2	-0.021	0.001	6	Confidence
CWomensMvt	-0.063	<0.001	2	0.049	0.001	5	Confidence
ILFam	-0.009	<0.001	3	0.009	0.001	6	Attribute
CArmedForces	0.058	<0.001	2	0.038	0.001	7	Confidence
ICQDetermination	0.015	<0.001	3	0.008	0.001	7	Attribute

CPress	-0.053	<0.001	2	-0.030	0.001	7	Confidence
PolDemoc	0.016	<0.001	3	-0.004	<0.001	5	Attribute
CCourts	-0.045	<0.001	2	0.003	<0.001	6	Confidence
ACTReligion	-0.029	<0.001	2	0.005	<0.001	5	Attribute
HardWork	0.057	<0.001	2	-0.007	<0.001	5	Attribute
ICQUnselfishness	-0.004	<0.001	3	0.012	0.001	7	Attribute
ILPolitics	0.007	<0.001	3	0.022	0.001	6	Attribute
PrivateState	0.010	<0.001	3	0.098	0.001	6	Attribute
CGovernment	0.009	<0.001	2	0.004	<0.001	6	Confidence
CEU	NA	NA	0	-0.010	<0.001	6	Confidence
PolScale	NA	NA	1	-0.007	<0.001	7	Attribute

```

trend_gt <- kable(trend_wide, digits = 3,
  caption = "Wave trend coefficients: SGP vs Others") |>
  kable_styling(full_width = FALSE)

# trend_wide rank by p-value
top_vars <- head(trend_wide$Variable[!is.na(trend_wide$slope_SGP)], 6)

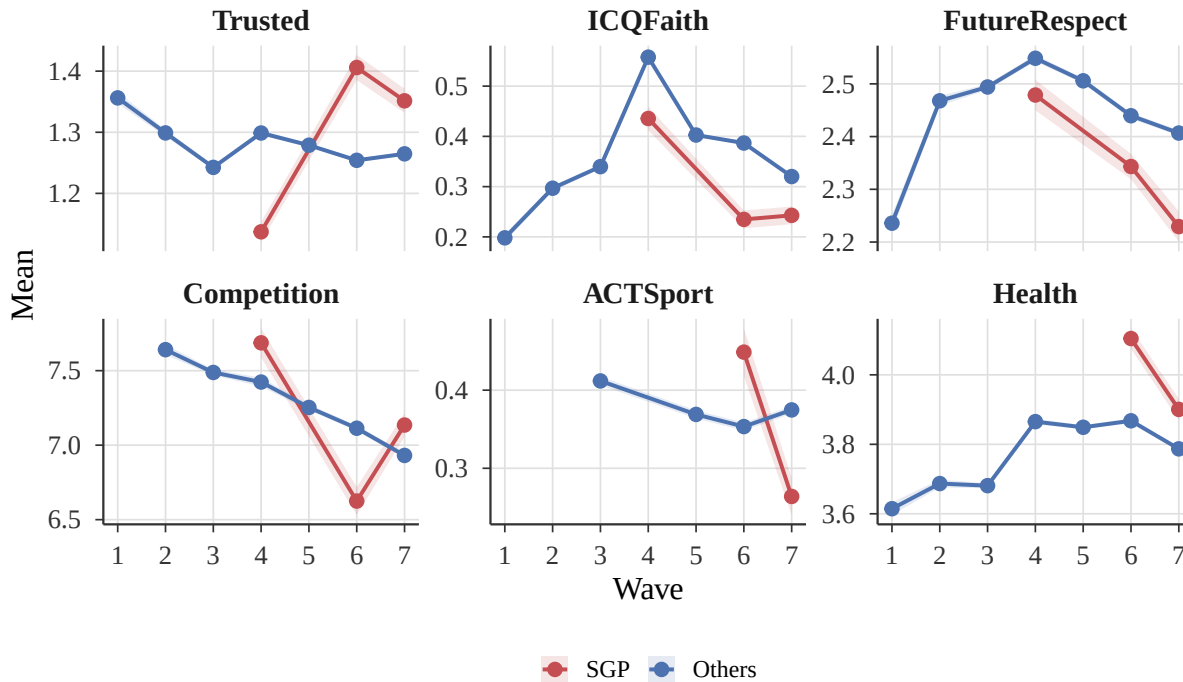
plot_trend <- wm_all[wm_all$Variable %in% top_vars, ]
plot_trend$Variable <- factor(plot_trend$Variable, levels = top_vars)

ggplot(plot_trend, aes(x = Wave, y = Mean, color = Group, group = Group)) +
  geom_ribbon(aes(ymin = Mean - SE, ymax = Mean + SE, fill = Group),
    alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 2) +
  facet_wrap(~ Variable, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("SGP" = "#C44E52", "Others" = "#4C72B0")) +
  scale_fill_manual(values = c("SGP" = "#C44E52", "Others" = "#4C72B0")) +
  scale_x_continuous(breaks = 1:7) +
  labs(title = "Temporal Shifts: SGP vs Others",
    subtitle = "Six variables with the strongest SGP wave trend",
    x = "Wave", y = "Mean", color = NULL, fill = NULL) +
  theme_pub

```

Temporal Shifts: SGP vs Others

Six variables with the strongest SGP wave trend



```
top_variables_change <- ggplot(plot_trend, aes(x = Wave, y = Mean, color = Group, group = Group)) +
  geom_ribbon(aes(ymin = Mean - SE, ymax = Mean + SE, fill = Group),
    alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 2) +
  facet_wrap(~ Variable, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("SGP" = "#C44E52", "Others" = "#4C72B0")) +
  scale_fill_manual(values = c("SGP" = "#C44E52", "Others" = "#4C72B0")) +
  scale_x_continuous(breaks = 1:7) +
  labs(title = "Temporal Shifts: SGP vs Others",
    subtitle = "Six variables with the strongest SGP wave trend",
    x = "Wave", y = "Mean", color = NULL, fill = NULL) +
  theme_pub

plot_trend[, c("Group", "Variable", "Wave", "Mean", "SD", "N", "SE")]
```

##	Group	Variable	Wave	Mean	SD	N
## ICQFaith_4	SGP	ICQFaith	4	0.4355556	0.4963814	450
## ICQFaith_6	SGP	ICQFaith	6	0.2347826	0.4242319	575
## ICQFaith_7	SGP	ICQFaith	7	0.2428331	0.4291570	593
## ACTSport_6	SGP	ACTSport	6	0.4486957	0.7351746	575
## ACTSport_7	SGP	ACTSport	7	0.2639594	0.5775787	591
## Health_6	SGP	Health	6	4.1045296	0.7135295	574

## Health_7	SGP	Health	7	3.9005059	0.7501462	593
## Trusted_4	SGP	Trusted	4	1.1370787	0.3443175	445
## Trusted_6	SGP	Trusted	6	1.4059233	0.4914982	574
## Trusted_7	SGP	Trusted	7	1.3514431	0.4778268	589
## FutureRespect_4	SGP	FutureRespect	4	2.4788419	0.6124617	449
## FutureRespect_6	SGP	FutureRespect	6	2.3432056	0.5749095	574
## FutureRespect_7	SGP	FutureRespect	7	2.2293103	0.6960308	580
## Competition_4	SGP	Competition	4	7.6868009	2.0803114	447
## Competition_6	SGP	Competition	6	6.6247818	2.2401018	573
## Competition_7	SGP	Competition	7	7.1351351	1.9175868	592
## ICQFaith_1	Others	ICQFaith	1	0.1981020	0.3986483	2529
## ICQFaith_2	Others	ICQFaith	2	0.2967927	0.4568788	6641
## ICQFaith_3	Others	ICQFaith	3	0.3397828	0.4736487	17676
## ICQFaith_41	Others	ICQFaith	4	0.5576460	0.4966845	13288
## ICQFaith_5	Others	ICQFaith	5	0.4026692	0.4904483	18807
## ICQFaith_61	Others	ICQFaith	6	0.3867477	0.4870186	17929
## ICQFaith_71	Others	ICQFaith	7	0.3200378	0.4665023	20107
## ACTSport_3	Others	ACTSport	3	0.4118164	0.7216239	17078
## ACTSport_5	Others	ACTSport	5	0.3689526	0.7009435	17921
## ACTSport_61	Others	ACTSport	6	0.3533987	0.6892521	17742
## ACTSport_71	Others	ACTSport	7	0.3747323	0.6967175	20081
## Health_1	Others	Health	1	3.6146238	0.9648513	2831
## Health_2	Others	Health	2	3.6870018	0.9124530	6524
## Health_3	Others	Health	3	3.6809867	0.9435883	16783
## Health_4	Others	Health	4	3.8652669	0.8716971	12907
## Health_5	Others	Health	5	3.8490425	0.8483920	18747
## Health_61	Others	Health	6	3.8676858	0.8622943	17859
## Health_71	Others	Health	7	3.7869095	0.8751426	20259
## Trusted_1	Others	Trusted	1	1.3562407	0.4789765	2692
## Trusted_2	Others	Trusted	2	1.2989274	0.4578235	6433
## Trusted_3	Others	Trusted	3	1.2425326	0.4286270	17342
## Trusted_41	Others	Trusted	4	1.2987791	0.4577404	13023
## Trusted_5	Others	Trusted	5	1.2787956	0.4484197	17303
## Trusted_61	Others	Trusted	6	1.2541092	0.4353717	17461
## Trusted_71	Others	Trusted	7	1.2647162	0.4411930	20029
## FutureRespect_1	Others	FutureRespect	1	2.2357221	0.8413556	2749
## FutureRespect_2	Others	FutureRespect	2	2.4678994	0.7736061	6760
## FutureRespect_3	Others	FutureRespect	3	2.4941119	0.6991655	16304
## FutureRespect_41	Others	FutureRespect	4	2.5485838	0.6912301	12463
## FutureRespect_5	Others	FutureRespect	5	2.5058003	0.6939380	17499
## FutureRespect_61	Others	FutureRespect	6	2.4395546	0.7182040	17065
## FutureRespect_71	Others	FutureRespect	7	2.4066755	0.7559335	19684
## Competition_2	Others	Competition	2	7.6409759	2.6247864	6189
## Competition_3	Others	Competition	3	7.4878826	2.5183924	15061
## Competition_41	Others	Competition	4	7.4235586	2.6618521	8065
## Competition_5	Others	Competition	5	7.2532754	2.4560658	17250
## Competition_61	Others	Competition	6	7.1135242	2.5525495	17221
## Competition_71	Others	Competition	7	6.9310241	2.7112112	19949
##		SE				
## ICQFaith_4				0.023399642		
## ICQFaith_6				0.017691695		
## ICQFaith_7				0.017623366		
## ACTSport_6				0.030658902		
## ACTSport_7				0.023758414		

```

## Health_6      0.029782144
## Health_7      0.030804813
## Trusted_4     0.016322213
## Trusted_6     0.020514737
## Trusted_7     0.019688512
## FutureRespect_4 0.028903856
## FutureRespect_6 0.023996259
## FutureRespect_7 0.028901108
## Competition_4  0.098395351
## Competition_6  0.093581596
## Competition_7  0.078812331
## ICQFaith_1     0.007927121
## ICQFaith_2     0.005606402
## ICQFaith_3     0.003562578
## ICQFaith_41    0.004308745
## ICQFaith_5     0.003576296
## ICQFaith_61    0.003637203
## ICQFaith_71    0.003289881
## ACTSport_3     0.005521949
## ACTSport_5     0.005236027
## ACTSport_61    0.005174600
## ACTSport_71    0.004916591
## Health_1       0.018133868
## Health_2       0.011296751
## Health_3       0.007283628
## Health_4       0.007672783
## Health_5       0.006196275
## Health_61      0.006452484
## Health_71      0.006148509
## Trusted_1      0.009231594
## Trusted_2      0.005708097
## Trusted_3      0.003254842
## Trusted_41     0.004011102
## Trusted_5      0.003408976
## Trusted_61     0.003294774
## Trusted_71     0.003117446
## FutureRespect_1 0.016046939
## FutureRespect_2 0.009409066
## FutureRespect_3 0.005475615
## FutureRespect_41 0.006191721
## FutureRespect_5 0.005245828
## FutureRespect_61 0.005497872
## FutureRespect_71 0.005387991
## Competition_2  0.033364431
## Competition_3  0.020520904
## Competition_41 0.029640242
## Competition_5  0.018700165
## Competition_61 0.019451137
## Competition_71 0.019195648

```

```
# Q3(b) - Regression per wave: SGP vs Others
```

```
attr_cols <- x_vars
```

```

sgp_conf_cols <- confidence_vars[sapply(sgp_data[, confidence_vars],
                                       function(x) sum(!is.na(x)) > 0)]

cat("SGP confidence variables with data:", sgp_conf_cols, "\n")

## SGP confidence variables with data: CArmedForces CPress CUnions CTelevision CGovernment CWomensMvt C

fit_reg_wave <- function(data, group_label, conf_set) {
  waves <- sort(unique(data$Wave))
  rows <- list()

  for (w in waves) {
    sub_all <- data[data$Wave == w, ]

    for (cc in conf_set) {
      # Skip if confidence outcome has too few observations this wave
      if (sum(!is.na(sub_all[[cc]])) < 80) next

      # Only attributes that have data in this wave
      attr_ok <- attr_cols[sapply(sub_all[, attr_cols],
                                function(x) sum(!is.na(x)) > 0)]
      if (length(attr_ok) < 2) next

      sub <- sub_all[, c(cc, attr_ok)]
      sub <- sub[complete.cases(sub), ]

      if (nrow(sub) < 80 || sd(sub[[cc]]) == 0) next

      sd_x <- sapply(sub[, attr_ok, drop = FALSE], sd)
      keep <- attr_ok[sd_x > 0]
      if (length(keep) < 2) next

      # Standardise
      z <- as.data.frame(scale(sub[, c(cc, keep)]))
      names(z)[1] <- "Y"

      fit <- tryCatch(lm(Y ~ ., data = z), error = function(e) NULL)
      if (is.null(fit)) next

      s <- summary(fit)
      cf <- s$coefficients[-1, , drop = FALSE] # drop intercept

      rows[[paste(w, cc, sep = "_")]] <- data.frame(
        Group = group_label,
        Wave = w,
        Confidence = cc,
        Attribute = rownames(cf),
        Beta = cf[, 1],
        t_value = cf[, 3],
        p = cf[, 4],
        N = nrow(sub),

```

```

    Adj_R2 = s$adj.r.squared,
    row.names = NULL
  )
}
}

do.call(rbind, rows)
}

# Run for SGP and Others
sgp_wr <- fit_reg_wave(sgp_data, "SGP", sgp_conf_cols)
oth_wr <- fit_reg_wave(oth_data, "Others", confidence_vars)
time_reg <- rbind(sgp_wr, oth_wr)
time_reg$Group <- factor(time_reg$Group, levels = c("SGP", "Others"))

# Adj R2 per wave × confidence
r2_tbl <- unique(time_reg[, c("Group", "Wave", "Confidence", "Adj_R2", "N")])
r2_wide <- reshape(r2_tbl[, c("Group", "Wave", "Confidence", "Adj_R2")],
  idvar = c("Confidence", "Wave"),
  timevar = "Group",
  direction = "wide")
r2_wide <- r2_wide[order(r2_wide$Confidence, r2_wide$Wave), ]
rownames(r2_wide) <- NULL

kable(r2_wide, digits = 3,
  caption = "Adj R2 per wave && confidence outcome (SGP vs Others)" |>
  kable_styling(full_width = FALSE))

```

Table 7: Adj R² per wave & confidence outcome (SGP vs Others)

Wave	Confidence	Adj_R2.SGP	Adj_R2.Others
1	CArmedForces	NA	0.106
2	CArmedForces	NA	0.084
3	CArmedForces	NA	0.077
4	CArmedForces	NA	0.055
5	CArmedForces	NA	0.125
6	CArmedForces	0.115	0.085
7	CArmedForces	0.108	0.087
1	CCourts	NA	0.025
2	CCourts	NA	0.097
3	CCourts	NA	0.061
5	CCourts	NA	0.087
6	CCourts	0.033	0.064
7	CCourts	0.113	0.076
2	CEU	NA	0.010
3	CEU	NA	0.047
4	CEU	NA	0.101
5	CEU	NA	0.050
6	CEU	NA	0.033
7	CEU	NA	0.071

2	CGovernment	NA	0.037
3	CGovernment	NA	0.061
4	CGovernment	NA	0.076
5	CGovernment	NA	0.119
6	CGovernment	0.049	0.077
7	CGovernment	0.183	0.090
1	CPress	NA	0.014
2	CPress	NA	0.032
3	CPress	NA	0.048
4	CPress	NA	0.048
5	CPress	NA	0.065
6	CPress	0.074	0.054
7	CPress	0.105	0.050
2	CTelevision	NA	0.032
3	CTelevision	NA	0.047
4	CTelevision	NA	0.044
5	CTelevision	NA	0.078
6	CTelevision	0.087	0.060
7	CTelevision	0.135	0.047
1	CUnions	NA	0.011
2	CUnions	NA	0.066
3	CUnions	NA	0.070
4	CUnions	NA	0.046
5	CUnions	NA	0.089
6	CUnions	0.077	0.079
7	CUnions	0.084	0.093
3	CWomensMvt	NA	0.032
4	CWomensMvt	NA	0.022
5	CWomensMvt	NA	0.052
6	CWomensMvt	0.084	0.051
7	CWomensMvt	0.032	0.069

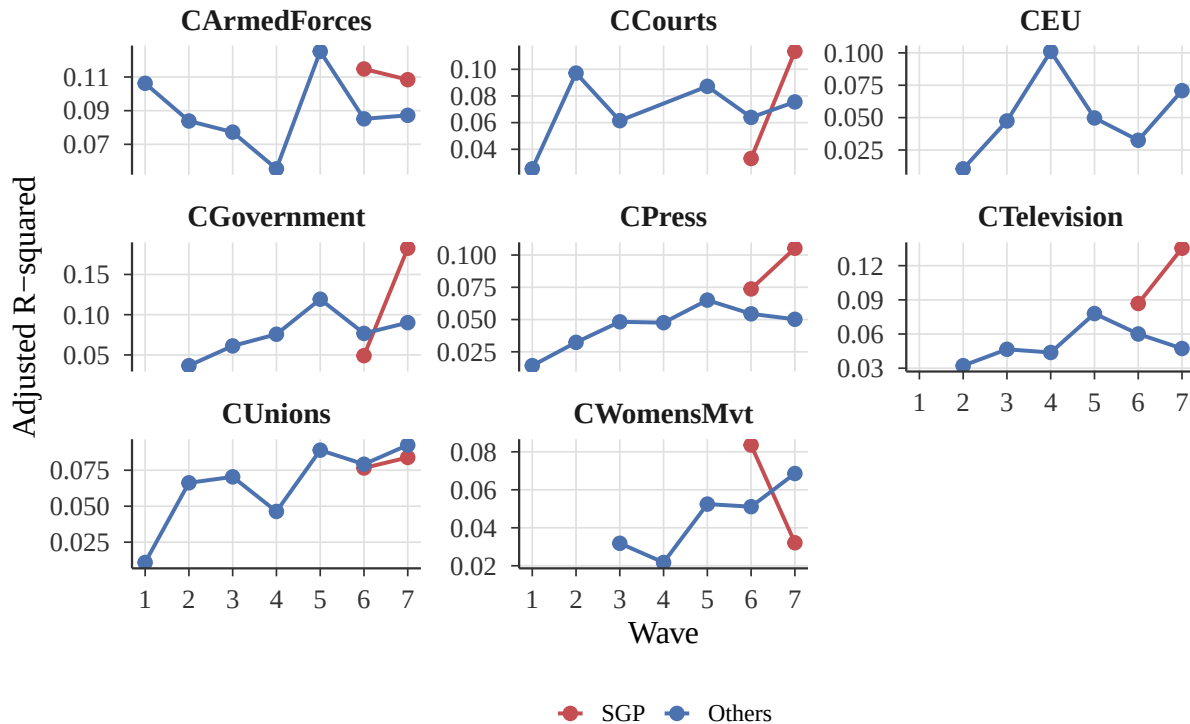
```

Q3B_Pre_Power <- ggplot(r2_tbl, aes(x = as.numeric(as.character(Wave)), y = Adj_R2,
                                   color = Group, group = Group)) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 2) +
  facet_wrap(~ Confidence, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("SGP" = "#C44E52", "Others" = "#4C72B0")) +
  scale_x_continuous(breaks = 1:7) +
  labs(title = "Predictive Power of Attributes over Waves",
       x = "Wave", y = "Adjusted R-squared", color = NULL) +
  theme_pub

print(Q3B_Pre_Power)

```

Predictive Power of Attributes over Waves



```
# Top predictors per wave × confidence
top_pool <- aggregate(abs(t_value) ~ Attribute, data = time_reg,
                      FUN = mean, na.rm = TRUE)

names(top_pool)[2] <- "Imp"
top_pool <- top_pool[order(-top_pool$Imp), ]
top_attrs <- head(top_pool$Attribute, 10)

heat_dat <- time_reg[time_reg$Attribute %in% top_attrs, ]
heat_dat <- aggregate(t_value ~ Group + Wave + Attribute,
                      data = heat_dat, FUN = mean, na.rm = TRUE)
heat_dat$Attribute <- factor(heat_dat$Attribute, levels = rev(top_attrs))
heat_dat$Group <- factor(heat_dat$Group, levels = c("SGP", "Others"))

Q3B_Pre_Strength <- ggplot(heat_dat, aes(x = factor(Wave), y = Attribute, fill = t_value)) +
  geom_tile(color = "white", linewidth = 0.2) +
  facet_wrap(~ Group, scales = "free_x", ncol = 2) +
  scale_fill_gradient2(low = "#2166AC", mid = "white", high = "#B2182B",
                      midpoint = 0) +
  labs(title = "Top Predictor Strength across Waves",
       subtitle = "Mean t-statistic across confidence variables",
       x = "Wave", y = NULL, fill = "Mean t") +
  theme_pub +
  theme(legend.position = "right",
       legend.direction = "vertical",
       legend.key.height = unit(1.2, "lines"))
```



```
ggsave("Q3B_Pre_Strength.png", plot = Q3B_Pre_Strength, width = 10, height = 6, dpi = 300)
print(Q3B_Pre_Strength)
```

Top Predictor Strength across Waves

Mean t-statistic across confidence variables



```
r2_tbl[order(r2_tbl$Group, r2_tbl$Confidence, r2_tbl$Wave), ]
```

##	Group	Wave	Confidence	Adj_R2	N
## 6_CArmedForces.1	SGP	6	CArmedForces	0.11477192	572
## 7_CArmedForces.1	SGP	7	CArmedForces	0.10841280	470
## 6_CCourts.1	SGP	6	CCourts	0.03308685	572
## 7_CCourts.1	SGP	7	CCourts	0.11340927	467
## 6_CGovernment.1	SGP	6	CGovernment	0.04910787	572
## 7_CGovernment.1	SGP	7	CGovernment	0.18256126	469
## 6_CPress.1	SGP	6	CPress	0.07372292	572
## 7_CPress.1	SGP	7	CPress	0.10526515	468
## 6_CTelevision.1	SGP	6	CTelevision	0.08678811	572
## 7_CTelevision.1	SGP	7	CTelevision	0.13530642	470
## 6_CUnions.1	SGP	6	CUnions	0.07656769	572
## 7_CUnions.1	SGP	7	CUnions	0.08388706	447
## 6_CWomensMvt.1	SGP	6	CWomensMvt	0.08354944	571
## 7_CWomensMvt.1	SGP	7	CWomensMvt	0.03210501	400
## 1_CArmedForces.1	Others	1	CArmedForces	0.10625402	1246
## 2_CArmedForces.1	Others	2	CArmedForces	0.08386484	3721
## 3_CArmedForces.1	Others	3	CArmedForces	0.07724215	8825
## 4_CArmedForces.1	Others	4	CArmedForces	0.05544203	4551

## 5_CARmedForces.1	Others	5	CARmedForces	0.12519578	7331
## 6_CARmedForces.110	Others	6	CARmedForces	0.08513582	11428
## 7_CARmedForces.110	Others	7	CARmedForces	0.08722108	11889
## 1_CCourts.1	Others	1	CCourts	0.02534905	1247
## 2_CCourts.1	Others	2	CCourts	0.09715924	3718
## 3_CCourts.1	Others	3	CCourts	0.06146494	8839
## 5_CCourts.1	Others	5	CCourts	0.08722176	7343
## 6_CCourts.110	Others	6	CCourts	0.06391936	11405
## 7_CCourts.110	Others	7	CCourts	0.07551226	11913
## 2_CEU.1	Others	2	CEU	0.01046590	1006
## 3_CEU.1	Others	3	CEU	0.04741725	4033
## 4_CEU.1	Others	4	CEU	0.10112352	774
## 5_CEU.1	Others	5	CEU	0.04974960	3338
## 6_CEU.1	Others	6	CEU	0.03250476	2568
## 7_CEU.1	Others	7	CEU	0.07084107	2676
## 2_CGovernment.1	Others	2	CGovernment	0.03666194	2052
## 3_CGovernment.1	Others	3	CGovernment	0.06119469	8809
## 4_CGovernment.1	Others	4	CGovernment	0.07567696	4532
## 5_CGovernment.1	Others	5	CGovernment	0.11925934	7345
## 6_CGovernment.110	Others	6	CGovernment	0.07672890	11431
## 7_CGovernment.110	Others	7	CGovernment	0.09012117	11959
## 1_CPress.1	Others	1	CPress	0.01427765	1248
## 2_CPress.1	Others	2	CPress	0.03233192	3710
## 3_CPress.1	Others	3	CPress	0.04824807	8866
## 4_CPress.1	Others	4	CPress	0.04757040	4576
## 5_CPress.1	Others	5	CPress	0.06506237	8062
## 6_CPress.110	Others	6	CPress	0.05442855	11442
## 7_CPress.110	Others	7	CPress	0.05020541	11946
## 2_CTelevision.1	Others	2	CTelevision	0.03223132	1635
## 3_CTelevision.1	Others	3	CTelevision	0.04659457	8861
## 4_CTelevision.1	Others	4	CTelevision	0.04377720	4582
## 5_CTelevision.1	Others	5	CTelevision	0.07796623	8081
## 6_CTelevision.110	Others	6	CTelevision	0.06003616	11455
## 7_CTelevision.110	Others	7	CTelevision	0.04727248	11957
## 1_CUnions.1	Others	1	CUnions	0.01089273	1245
## 2_CUnions.1	Others	2	CUnions	0.06626258	3663
## 3_CUnions.1	Others	3	CUnions	0.07041049	8518
## 4_CUnions.1	Others	4	CUnions	0.04634758	4394
## 5_CUnions.1	Others	5	CUnions	0.08896958	7828
## 6_CUnions.110	Others	6	CUnions	0.07918115	11089
## 7_CUnions.110	Others	7	CUnions	0.09250127	11647
## 3_CWomensMvt.1	Others	3	CWomensMvt	0.03182982	8275
## 4_CWomensMvt.1	Others	4	CWomensMvt	0.02173284	4270
## 5_CWomensMvt.1	Others	5	CWomensMvt	0.05246110	7741
## 6_CWomensMvt.110	Others	6	CWomensMvt	0.05110020	11052
## 7_CWomensMvt.110	Others	7	CWomensMvt	0.06856951	11524

```
print(Q3B_Pre_Strength)
```

Top Predictor Strength across Waves

Mean t-statistic across confidence variables



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.